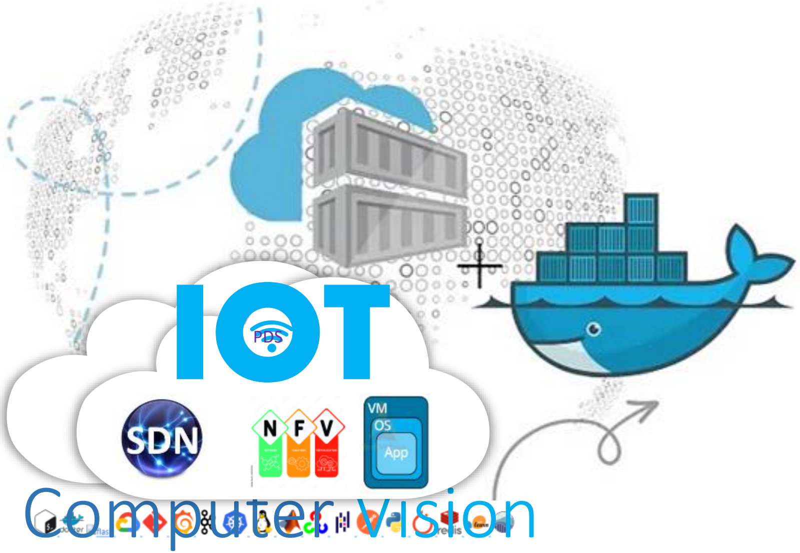

Kolla Ansible OpenStack Installation (Ubuntu server 22.04)

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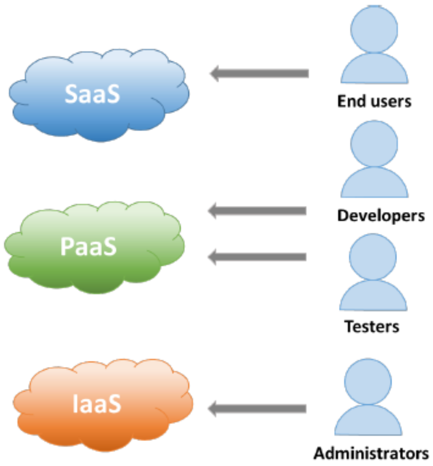
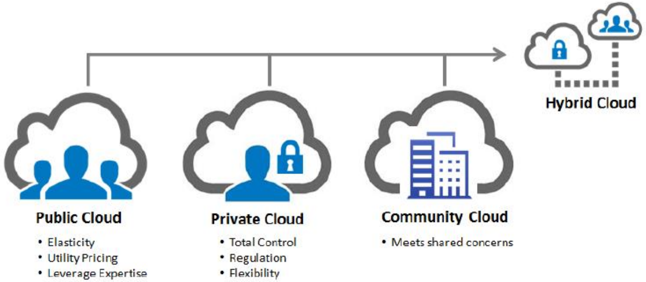
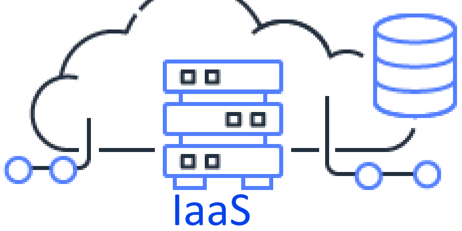
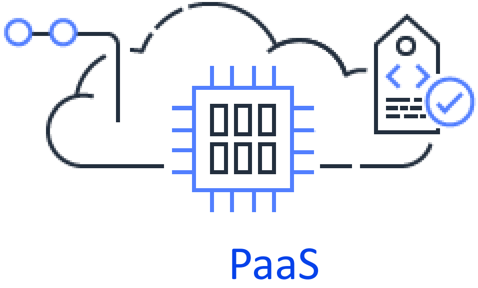
https://github.com/siagianp
https://www.youtube.com/@AmelOline/videos
https://github.com/amelcharolinesgn2/IoT_simulator-mqtt-NodeRed
https://github.com/amelcharolinesgn2/Cloud-Infrastructures
https://github.com/amelcharolinesgn2/ClouD-Infrastructure-SISKA-2025


SaaS
End users
PaaS
Developers
Testers
IaaS
Administrators

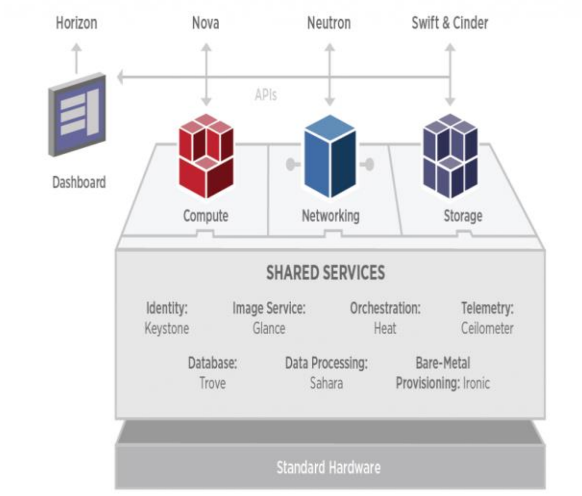

Hybrid Cloud
Public Cloud
• Elasticity
• Utility Pricing
• Leverage Expertise
Private Cloud
• Total Control
• Regulation
• Flexibility
Community Cloud
• Meets shared concerns

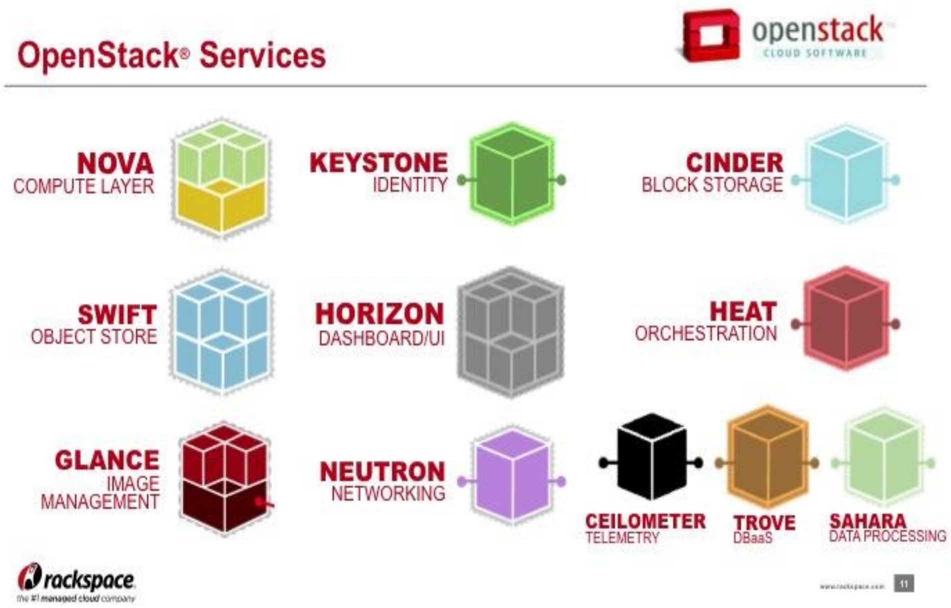

IaaS


PaaS


SaaS

Arsitektur Openstack





Openstack Services

Kolla Ansible melakukan deploy container untuk project Openstack berikut:

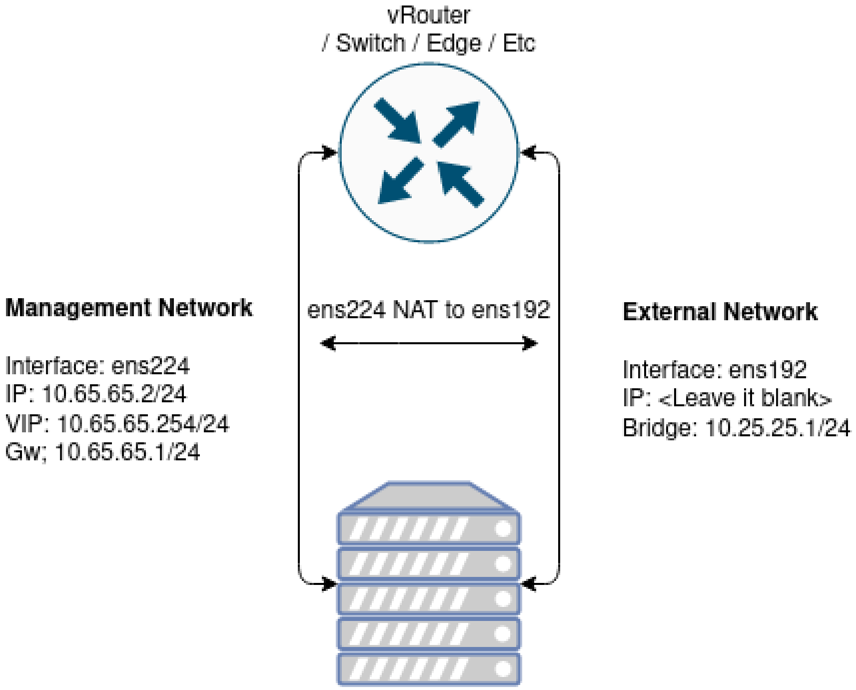
Aodh
Barbican
Bifrost
Blazar
Ceilometer
Cinder
CloudKitty
Congress
Designate
Freezer
Glance
Heat
Horizon
Ironic
Karbor
Keystone
Kuryr
Magnum
Manila
Mistral
Murano

Infrastructure components

Service Openstack berikut:

- ✓ Ceph implementation for Cinder, Glance and Nova.
- ✓ Collectd, Telegraf, InfluxDB, Prometheus, and Grafana for performance monitoring.
- ✓ Elasticsearch and Kibana to search, analyze, and visualize log messages.
- ✓ Etcd a distributed reliable key-value store.
- ✓ Fluentd as an open source data collector for unified logging layer.
- ✓ Gnocchi A time-series storage database.
- ✓ HAProxy and Keepalived for high availability of services and their endpoints.
- ✓ MariaDB and Galera Cluster for highly available MySQL databases.
- ✓ Memcached a distributed memory object caching system.
- ✓ MongoDB as a database back end for Panko.
- ✓ Open vSwitch and Linuxbridge backends for Neutron.
- ✓ RabbitMQ as a messaging backend for communication between services.
- ✓ Redis an in-memory data structure store.

Lab: Kolla Ansible All in One Openstack



Ubuntu 20.04
Openstack

Requirement

Kali ini kami akan menggunakan spesifikasi mesin dan os seperti berikut, silakan dapat diikuti atau dapat disesuaikan dengan kebutuhan Anda:

- ✓ 8 CPU
- ✓ 8 GB RAM
- ✓ 100 GB Disk
- ✓ Ubuntu 20.04
- ✓ 2 Network Interface

Lab: Kolla Ansible All in One Openstack (Cons't)

2 NIC

Membutuhkan minimal 2 network interface yang nantinya akan digunakan sebagai:

1. Management Network

Interface: ens224 IP: 10.65.65.2/24 VIP: 10.65.65.254/24 Gw; 10.65.65.1/24

2. External Network

Interface: ens192 IP: kosongkan Bridge: 10.25.25.1/24

Management Network digunakan sebagai media komunikasi antar service yang menjalankan openstack, sedangkan External Network digunakan service neutron untuk berkomunikasi antara network internal (private) pada instances dan host (internet/publik), untuk interface ini diharuskan berstatus aktif (up) dan tidak dialokasikan IP pada interfacenya.

Konfigurasi Network

Saran...!!!!
Menggunakan ip statis pada setiap interface untuk menghindari perubahan IP address / routing pada saat host restart.
Network manager saya menggunakan netplan, berikut adalah konfigurasi interfacenya:

```
vi /etc/netplan/01-iface.yaml
...
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens224:
      dhcp4: false
      addresses: [10.65.65.2/24]
      nameservers:
         addresses: [1.1.1.1,1.0.0.1]
    ens192: {}
  version: 2
  ...
    netplan apply
```

Instalasi dependensi

Menggunakan media virtual environment yang disediakan python, adapun dependensi yang perlu diinstall adalah berikut:

root@administrator:/home/panda/Deploy#

Update paket manager:

 # sudo apt update

Install dependensi python3-venv:

 # sudo apt install python3-venv

Buat sebuah virtual environment lalu aktifkan untuk mulai menggunakannya:

- Buat folder baru pada direktori home ***panda*** denga nama **Deploy**, bisa disesuaikan dengan fungsinya :)
- mengaktifkan virtual environment dahulu sebelum memulai menggunakan perintah python

```
mkdir deploy
python3 -m venv /home/panda/Deploy
source /home/panda/Deploy/bin/activate
```

Upgrade versi pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

root@administrator:/home/panda/Deploy#

 # pip install –U pip

Install paket ansible :

Kolla ansible membutuhkan paket ansible minimal versi 4 dan mendukung hingga versi 5 keatas.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install 'ansible>=4,<6'
```

Lab: Kolla Ansible All in One Openstack (Cons't)

root@administrator:/home/panda/Deploy#

Install paket docker:

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install docker
```

Install Kolla-ansible

Install kolla-ansible beserta dependensinya menggunakan python-pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
#pip install git+https://opendev.org/openstack/kolla-ansible@master
```

Buat directory /etc/kola

root@administrator:/home/panda/Deploy#

```
# sudo mkdir -p /etc/kolla
# sudo chown $panda:$panda /etc/kolla
```

Salin file global.yml dan password.yml ke directory /etc/kolla:

```
# cp -r /home/panda/deploy/share/kolla-ansible/etc_examples/kolla/* /etc/kolla
```

Salin file konfigurasi all-in-one:

```
# cp /path/to/venv/share/kolla-ansible/ansible/inventory/all-in-one /home/panda/
```

Tambahkan python interpreter pada konfigurasi all-in-one

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# source /home/panda/deploy/bin/activate
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Cari lokasi binary python:

```
which python
/home/panda/Deploy/bin/python
```

Tambahkan interpreter sesuai lokasi

```
vi all-in-one
...
Localhost
ansible_python_interpreter=/home/panda/deploy/bin/python
```

Install Ansible Galaxy requirements

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-ansible install-deps
```

Konfigurasi Awal

Generate Password

Password yang digunakan oleh kolla ansible disimpan di direktori **/etc/kolla/passwords.yml**

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-genpwd
```

Konfigurasi global.yml

```
vi /etc/kolla/global.yml
...
kolla_base_distro: "ubuntu"
openstack_release: "master"
kolla_internal_vip_address: "10.65.65.254"
network_interface: "ens224"
neutron_external_interface: "ens192"
neutron_plugin_agent: "openvswitch"
enable_neutron_provider_networks: "yes"
nova_compute_virt_type: "kvm"
```

Simpan konfigurasi dan lanjut ke deployment !

Lab: Kolla Ansible All in One Openstack (Cons't)

Deployment

Kolla Ansible sudah menyediakan playbook list yang akan menginstal semua service yang sudah di set pada konfigurasi sebelumnya.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

@ Lakukan bootstraping pada host menggunakan dependensi kolla ansible:

```
# kolla-ansible -i all-in-one
bootstrap-servers
```

@ Lakukan pre-checks konfigurasi

```
# kolla-ansible -i all-in-one prechecks
```

@ Jika dari hasil check sudah tidak ada error, selanjutnya deploy server

```
# kolla-ansible -i all-in-one deploy
```

Proses deployment akan membutuhkan waktu yang agak lama, pastikan saja kalian sudah menjalankan **screen** bila menggunakan remote SSH.

Jika deployment berhasil output akhirnya kurang lebih seperti berikut:

@ Lakukan bootstraping pada host menggunakan dependensi kolla ansible:

```
...
PLAY RECAP
*****************************************
*****************************************
*****************************************
***************************
localhost              : ok=366  changed=72
unreachable=0   **failed=0**   skipped=245
rescued=0   ignored=0
```

Lab: Kolla Ansible All in One Openstack (Cons't)

List Running OpenStack Docker Containers

Untuk mengecek service docker yang berjalan silakan menggunakan command ini:

```
# docker ps
```

OpenStack Post Deployment

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

Install Openstack Client

```
# pip install python-openstackclient python-neutronclient python-glanceclient
```

Generate AdminRC

```
# kolla-ansible post-deploy
```

Gunakan AdminRC

```
# source /etc/kolla/admin-openrc.sh
```

Jalankan script untuk inisiasi awal

Secara default kolla ansible sudah menyiapkan script inisiasi awal yang digunakan untuk create network demo didalam openstack.

```
vim /home/panda/deploy/share/kolla-ansible/init-runonce
...
# This EXT_NET_CIDR is your public network,that you want to connect to the internet via.
ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}
EXT_NET_CIDR=${EXT_NET_CIDR:-'10.25.25.0/24'}
EXT_NET_RANGE=${EXT_NET_RANGE:-'start=10.25.25.4,end=10.25.25.253'}
EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-'10.25.25.1'}
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Eksekusi script

```
/home/panda/deploy/share/kolla-ansible/init-runonce
```

Aktifkan ipv4 forwarding

```
vi /etc/sysctl.conf
...
net.ipv4.ip_forward=1
net.ipv4.ip_nonlocal_bind=1
net.ipv6.ip_nonlocal_bind=1
net.unix.max_dgram_qlen=128
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.neigh.default.gc_thresh1=128
net.ipv4.neigh.default.gc_thresh2=28672
net.ipv4.neigh.default.gc_thresh3=32768
net.ipv6.neigh.default.gc_thresh1=128
net.ipv6.neigh.default.gc_thresh2=28672
net.ipv6.neigh.default.gc_thresh3=32768
```

Sysctl

```
# sysctl -p
```

Atur interface bridge

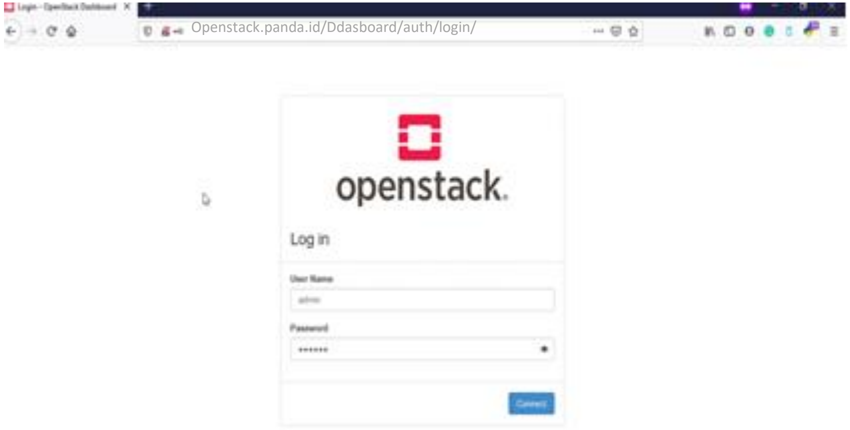
Jalankan perintah berikut ini :

```
export ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}
export EXT_NET_CIDR=${EXT_NET_CIDR:-'10.25.25.0/24'}
export EXT_NET_RANGE=${EXT_NET_RANGE:-'start=10.25.25.3,end=10.25.25.253'}
export EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-'10.25.25.1'}

## Akses internet ke instances
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask 255.255.255.0 up
sudo iptables -t nat -A POSTROUTING -s $EXT_NET_CIDR -o ens224 -j MASQUERADE
sudo iptables -A FORWARD -o ens224 -i br-ex -j ACCEPT
sudo iptables -A FORWARD -i ens224 -o br-ex -j ACCEPT
```

Akses dashboard dan uji konektifitas instance

Dashboard dapat diakses menggunakan Virtual IP yang sudah diset sebelumnya yaitu : http://10.65.65.254



Password dapat dilihat menggunakan command berikut ini:

```
# grep keystone_admin_password /etc/kolla/passwords.yml
```

Troubleshoot

Ketika host mengalami restart:

```
export EXT_NET_CIDR='10.25.25.0/24'
export EXT_NET_RANGE='start=10.25.25.3,end=10.25.25.253'
export EXT_NET_GATEWAY='10.25.25.1'
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask 255.255.255.0 up
sudo iptables -t nat -A POSTROUTING -s $EXT_NET_CIDR -o ens224 -j MASQUERADE
echo 1 > /proc/sys/net/ipv4/ip_forward
sudo iptables -A FORWARD -o ens224 -i br-ex -j ACCEPT
sudo iptables -A FORWARD -i ens224 -o br-ex -j ACCEPT
```

Membuat Instance Melalui Horizon

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

- ✓ **Openstack: Upload Image atau ISO**
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Upload Image atau ISO

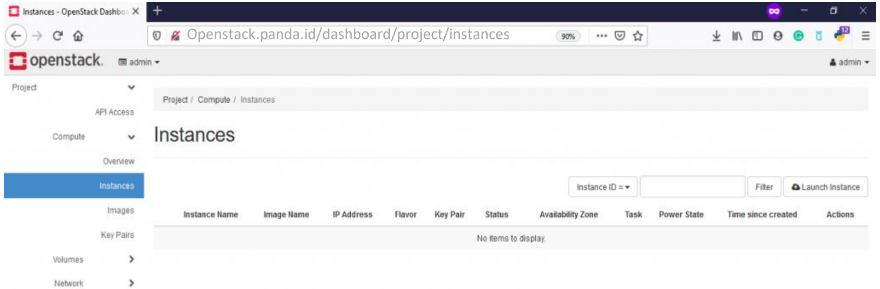
Create Instance

Lab: Kolla Ansible All in One Openstack (Cons't)

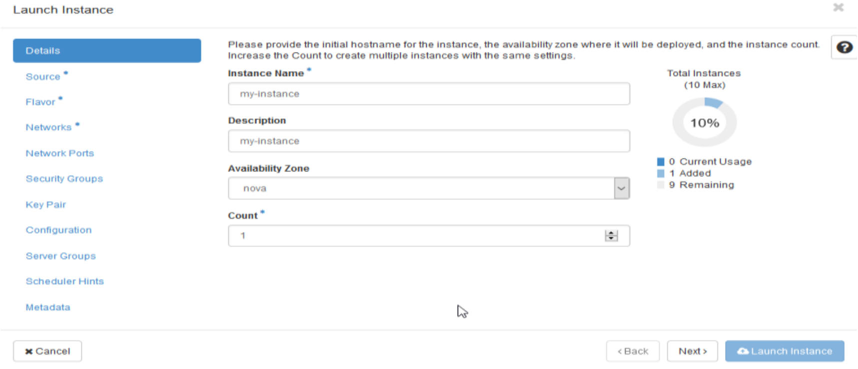
Upload Image atau ISO

Create Instance

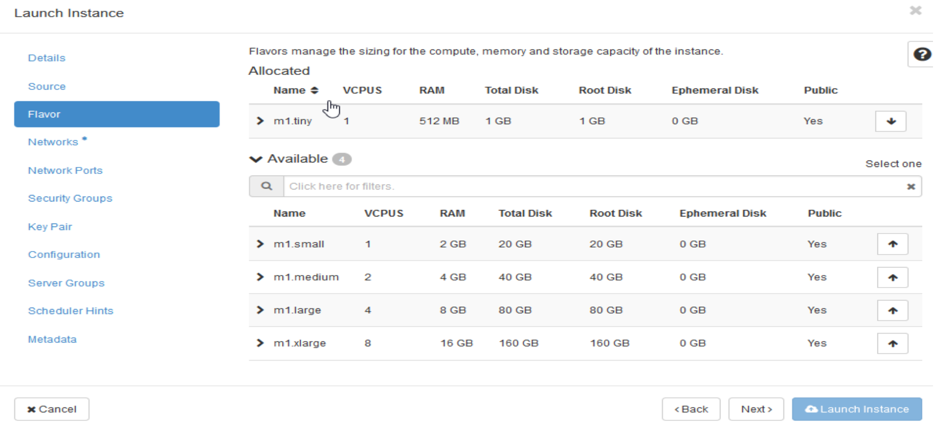
Upload image atau iso menggunakan cirros, silakan Login Horizon >> Project >> Compute >> Instances >> Launch Instance



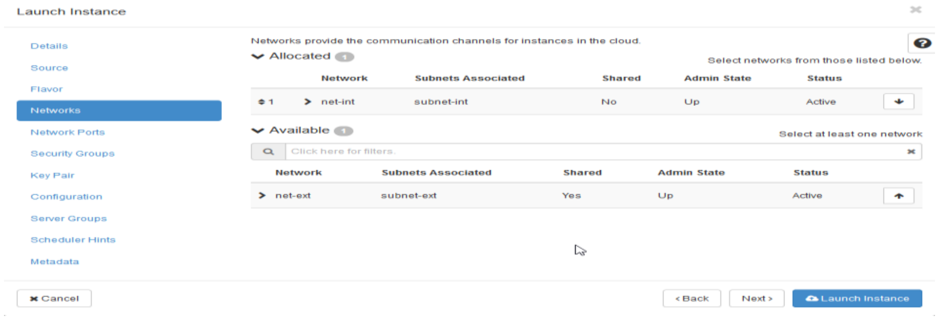
Pada menu Source silakan pilih OS atau image yang ingin Anda gunakan, disini kami contohkan OS yang kami gunakan yaitu Cirros



Menu **Flavor** silakan pilih spesifikasi instance



Menu **Networks** silakan pilih network internal yang telah dibuat sebelumnya.

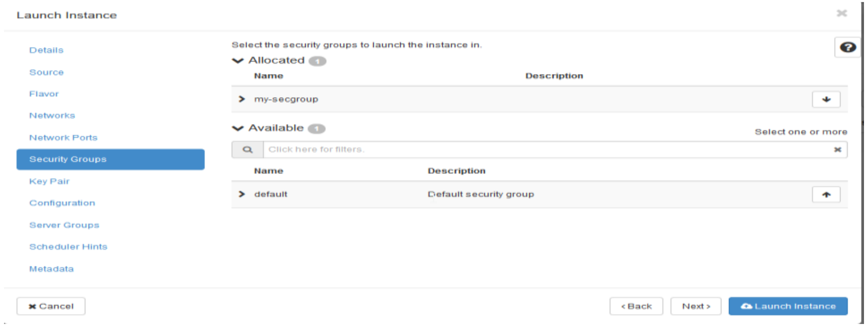


Lab: Kolla Ansible All in One Openstack (Cons't)

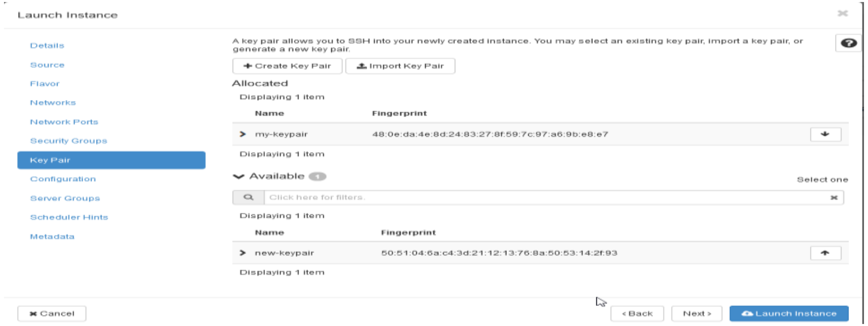
Upload Image atau ISO

Create Instance

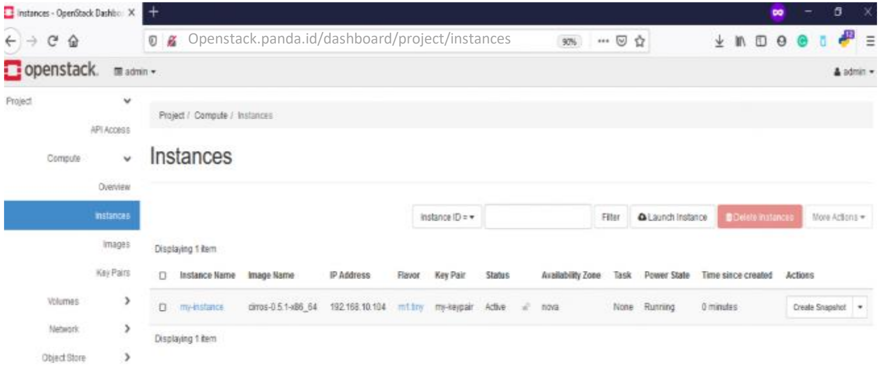
Menu Security Groups , silakan pilih Security Groups yang telah dibuat sebelumnya.



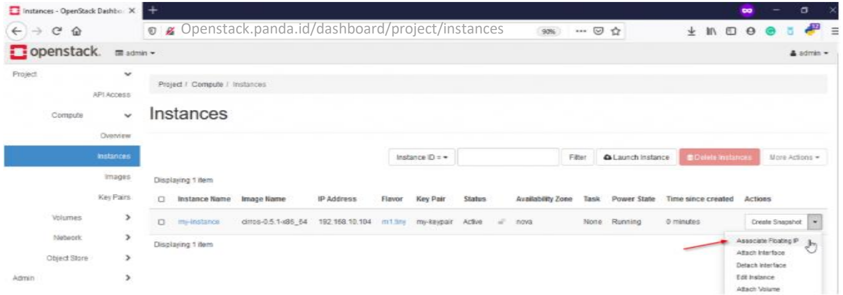
Menu **Key Pair** silakan pilih key pair yang telah di buat juga sebelumnya



Jika sudah silakan, create **Launch Instance** silakan tunggu proses pembuatan Instance selesai



Saat ini instance sudah mendapatkan IP internal secara otomatis, supaya dapat diakses dari public silakan membuat floating IP pada instance silakan klik menu Associate Floating IP



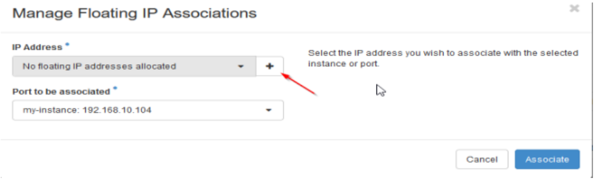
Klik tanda +

Lab: Kolla Ansible All in One Openstack (Cons't)

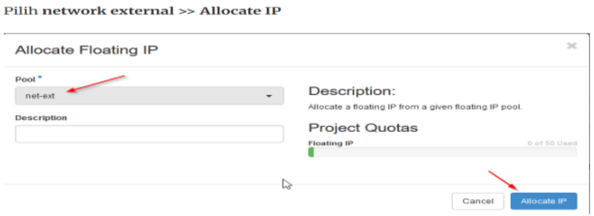
Upload Image atau ISO

Create Instance

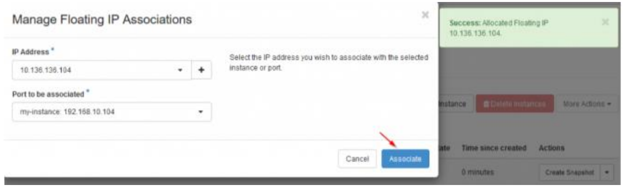
Manage Floating IP Associations



Pilih network external >> Allocate IP



Saat ini kita sudah mendapatkan IP nya, silakan klik **Associate** untuk dapat di assign ke instance Anda

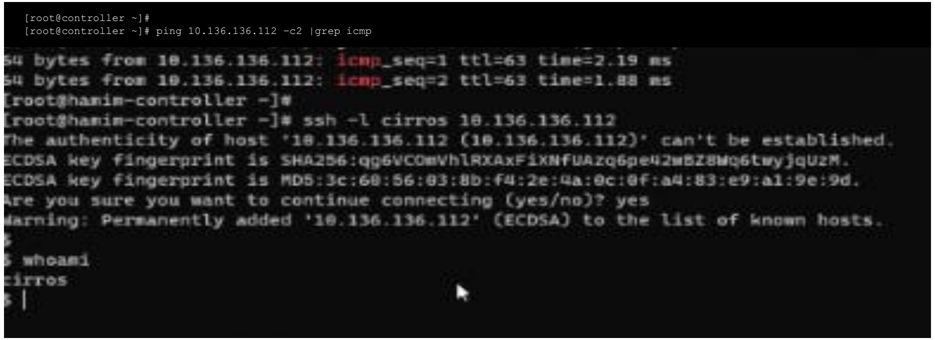


Saat ini instance sudah mendapatkan dan menggunakan IP dari network internal dan external dan silakan test ping ke IP floating

```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c4
```

Test login ke sisi instance

```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c2 |grep icmp
```



Upload Image atau ISO

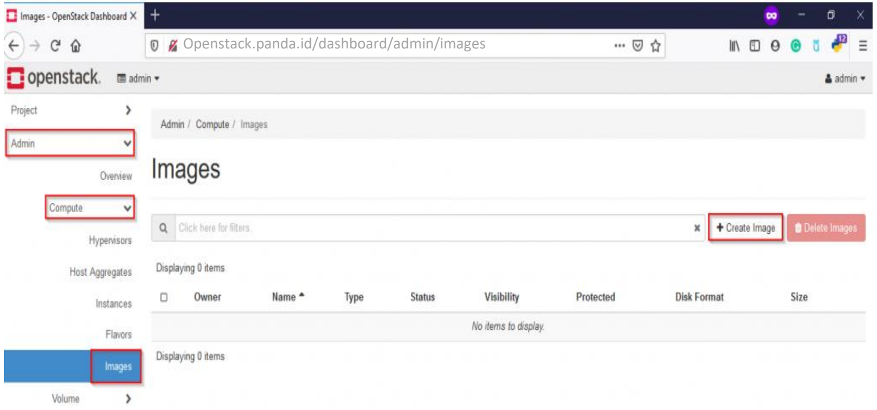
Create Image

upload image atau iso di openstack terdapat 2 cara yang dapat dilakukan yaitu melalui horizon (melalui web) atau CLI.

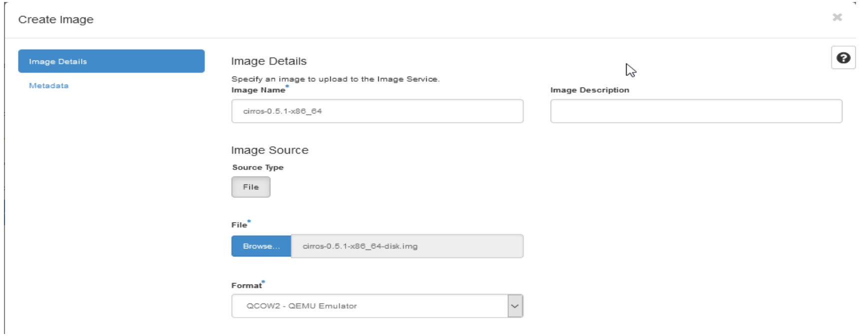
Upload image atau iso pastikan Anda sudah mengunguh image nya terlebih dahulu berikut link jika menggunakan Ubuntu, CentOS dan Cirros

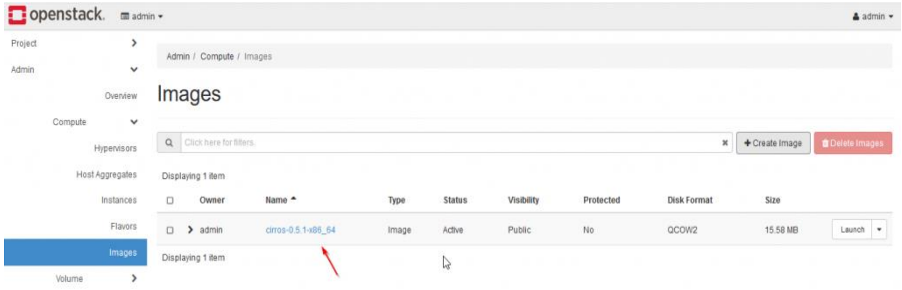
- Ubuntu Cloud Images
- CentOS Cloud Images
- Cirros

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



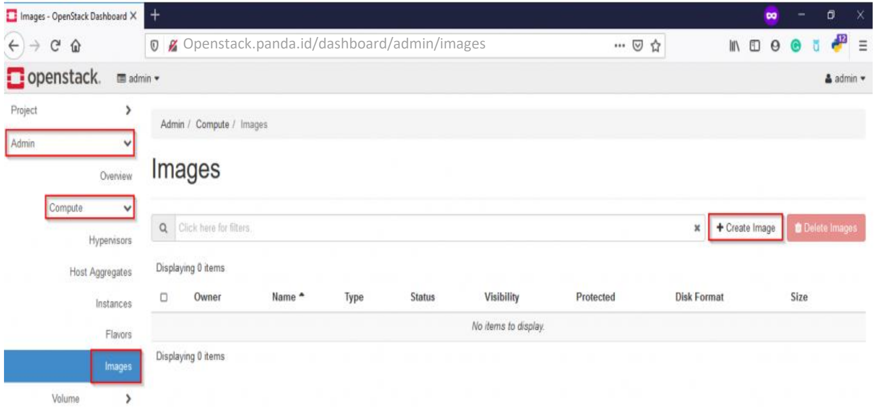
Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*



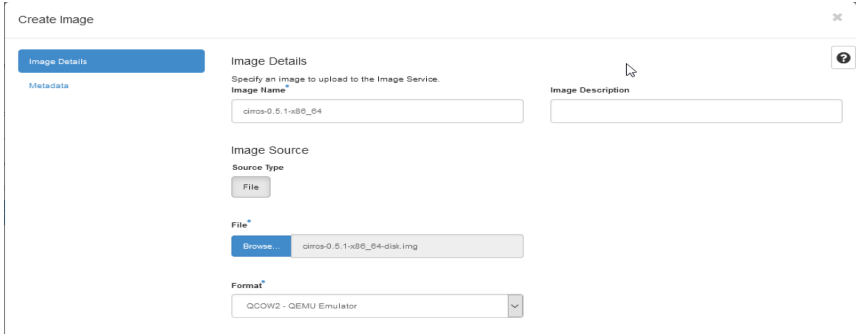


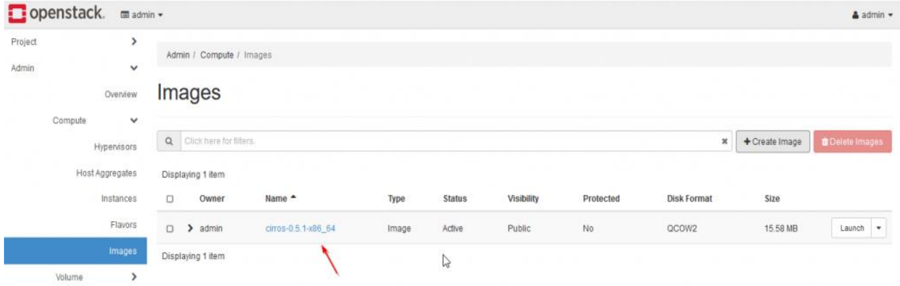
Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*

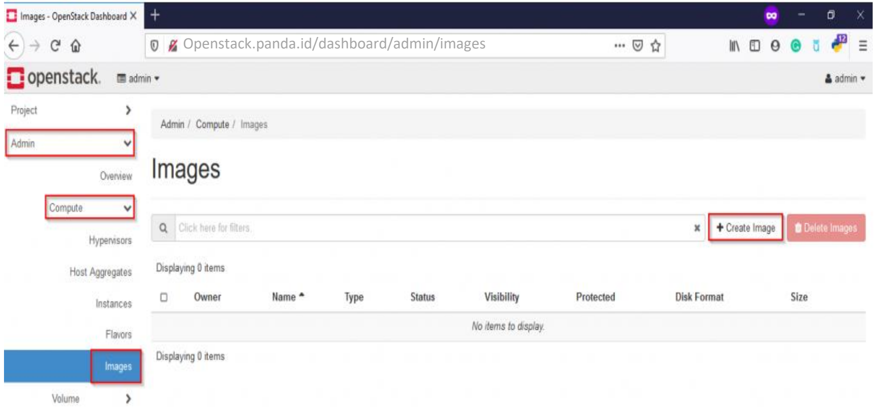




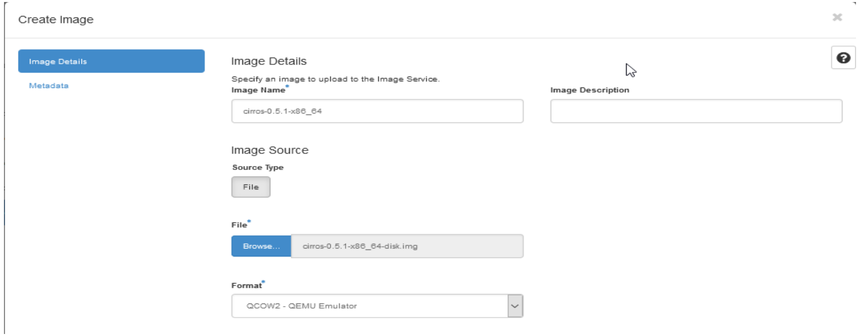
Lab: Kolla Ansible All in One Openstack (Cons't)

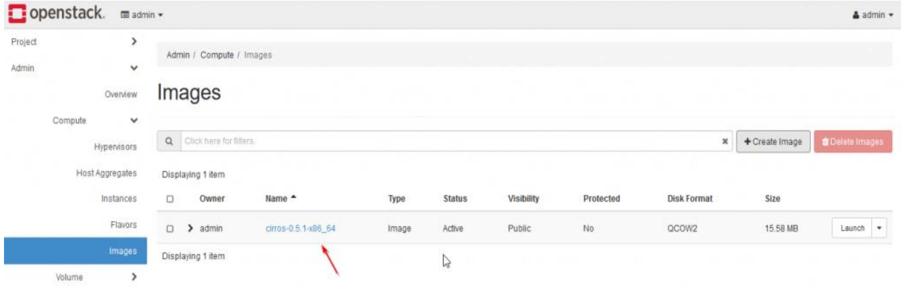
Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*





- ✓ **Openstack: Upload Image atau ISO**
- ✓ **Openstack: Membuat Network**
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Membuat Network

Create Network

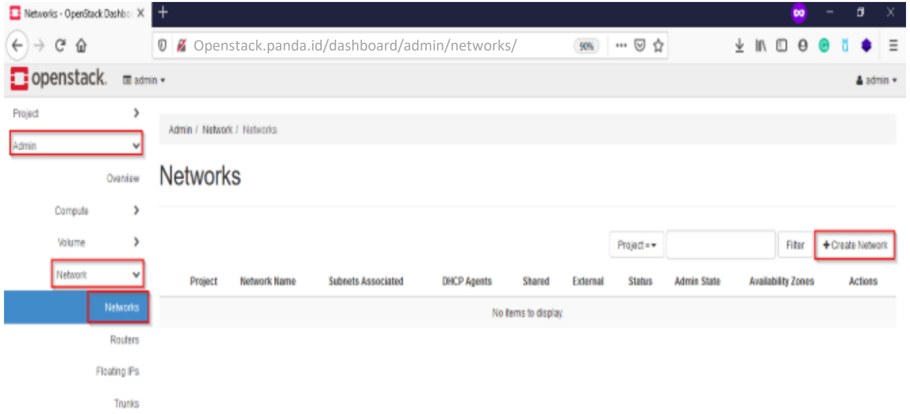
Create Network

Network yaitu network external dan network internal

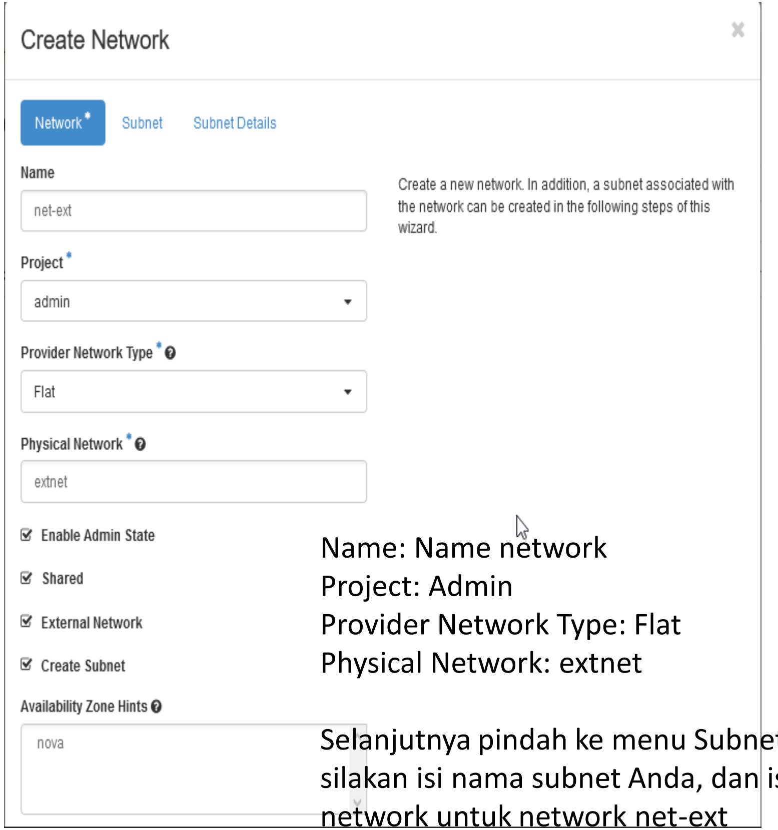
- Network External: Network yang tersambung keluar (node controller dan compute).
- Network internal: Network yang akan terhubung dan digunakan oleh instance/vm.

Login Horizon >> Admin >> Network >> Networks >> Create Network

Openstack.panda.id/dashboard/admin/networks/



Menu ***Network***

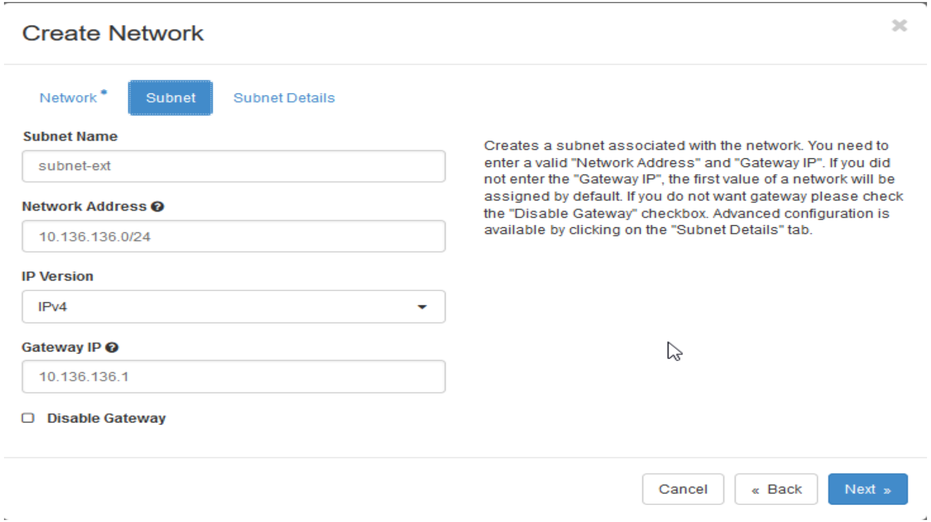


Name: Name network
Project: Admin
Provider Network Type: Flat
Physical Network: extnet

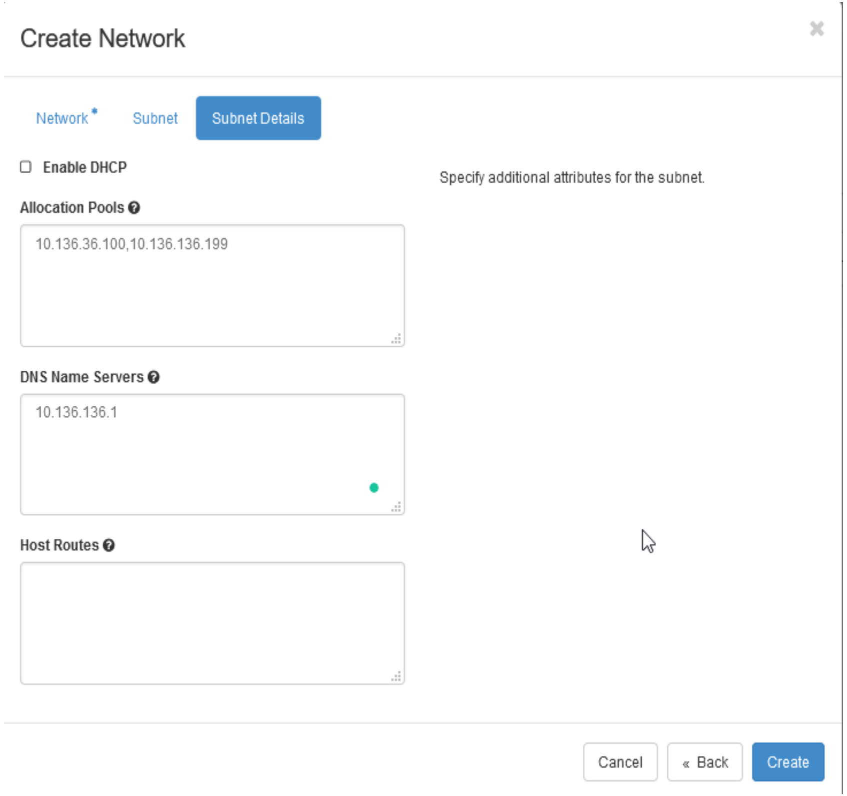
Selanjutnya pindah ke menu Subnet , silakan isi nama subnet Anda, dan isi juga network untuk network net-ext

Create Network

Menu Subnet , silakan isi nama subnet Anda, dan isi juga network untuk network net-ext

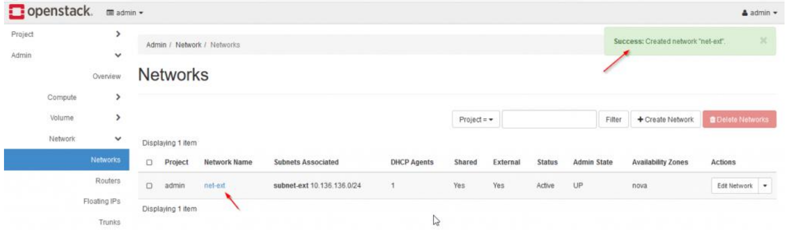


Menu _ Subnet Details _ silakan isi ip pool yang ingin Anda gunakan dari network address diatas, jika sudah silakan klik Create



Create Network

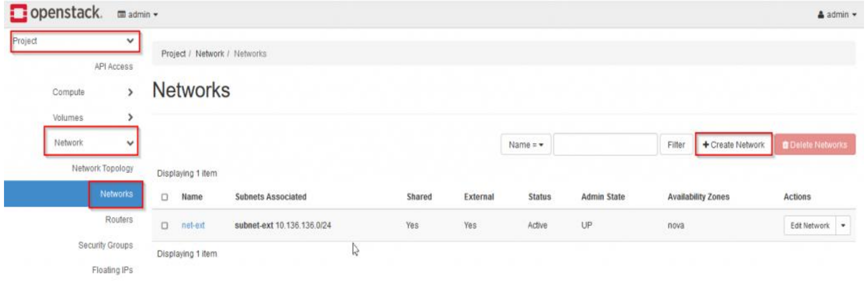
Pembuatan network selesai



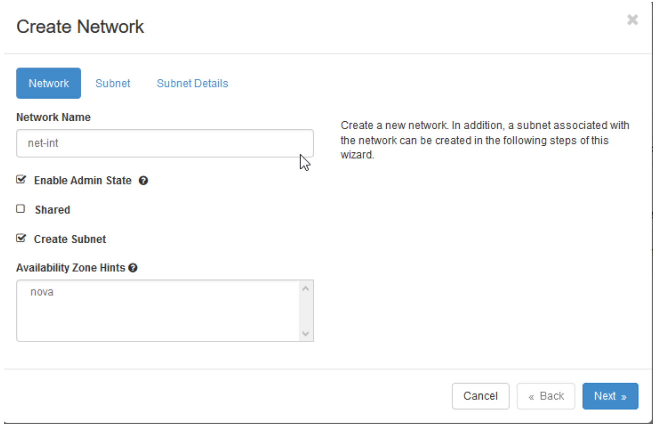
Saat ini external network sudah berhasil dibuat, selanjutnya create internal network seperti berikut:

Login Horizon >> Project >> Network >> Networks >> Create Network

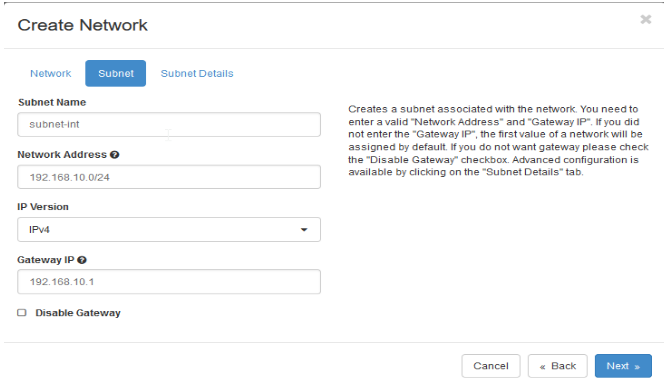
Openstack.panda.id/dashboard/admin/networks/Create Network



Menu Network isikan nama network

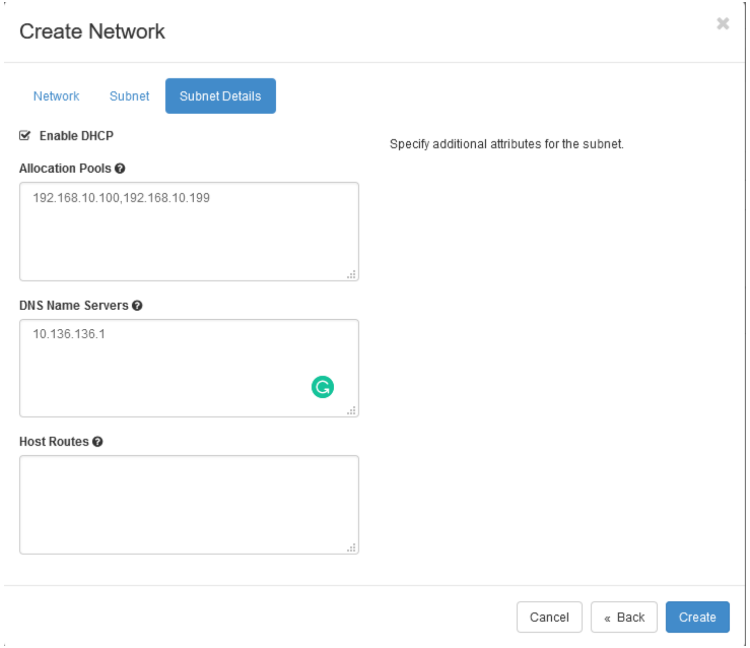


Pindah ke menu subnet untuk menentukan ip address dan subnet nya

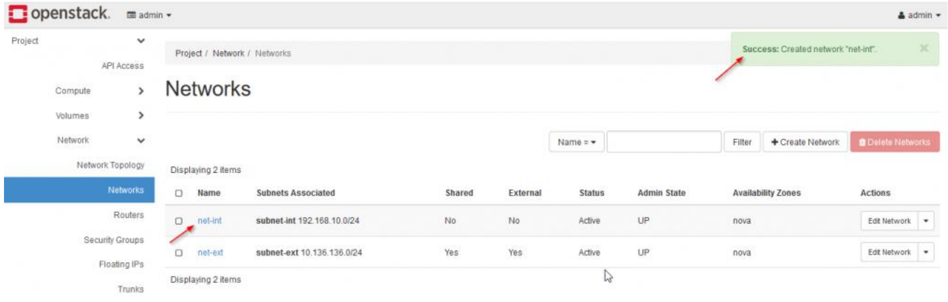


Create Network

Pada menu Subnet Details, silakan isi pool network yang ingin digunakan dari range ip address diatas lalu klik _ Create _ untuk membuat network internal



Proses pembuatan network internal



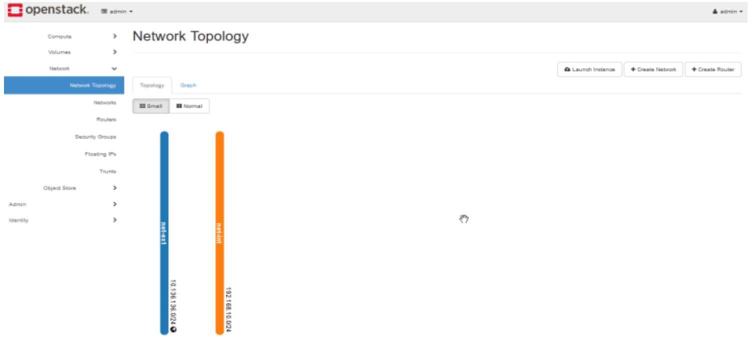
- ✓ **Openstack: Upload Image atau ISO**
- ✓ **Openstack: Membuat Network**
- ✓ **Openstack: Membuat Router**
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

- ✓ Membuat Router

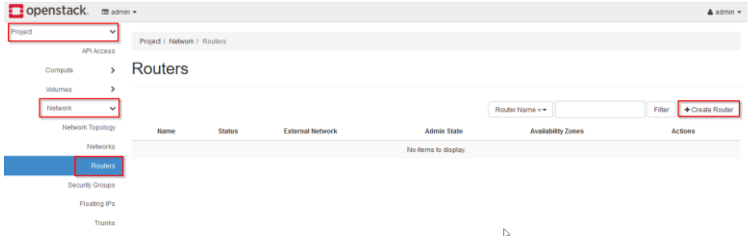
Create Router

Create Router

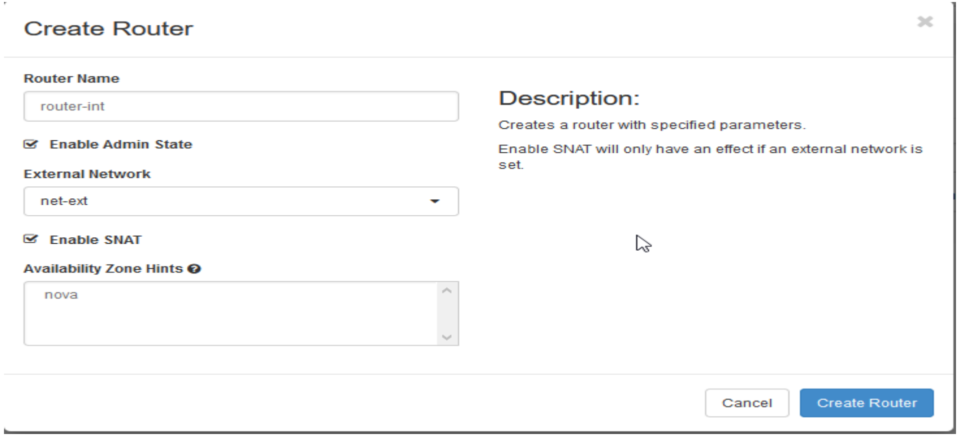
Membuat router melalui horizon, fungsi router sendiri untuk menghubungkan network yang telah kita buat sebelumnya, jika di lihat melalui *Network topologi* saat ini sudah terdapat 2 network internal dan external

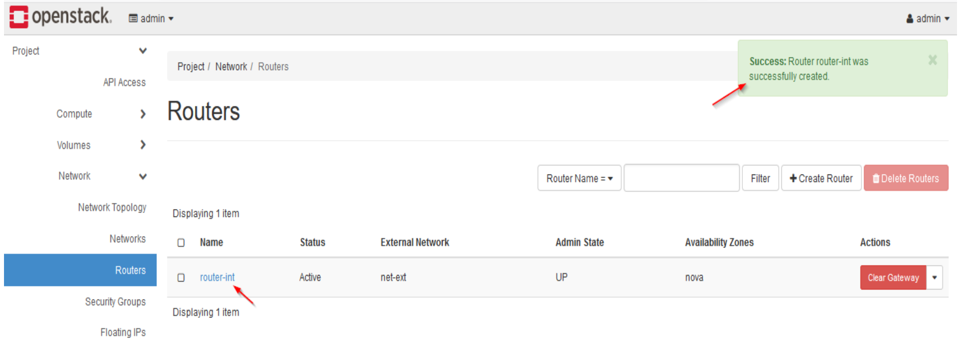


Membuat router Login ke Horizon >> Network >> Router >> Create Router



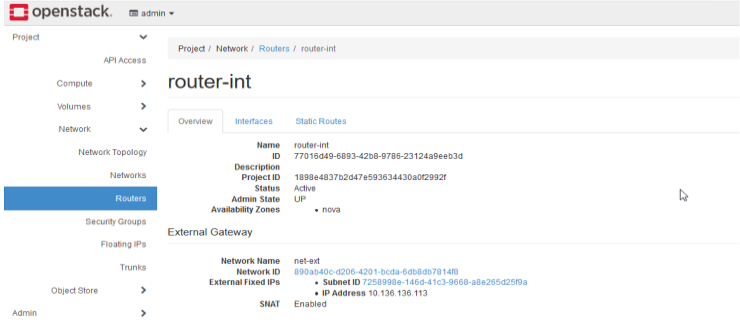
Isi nama Router yang Anda inginkan contoh seperti gambar dibawah ini, klik **Create Router** untuk membuat router



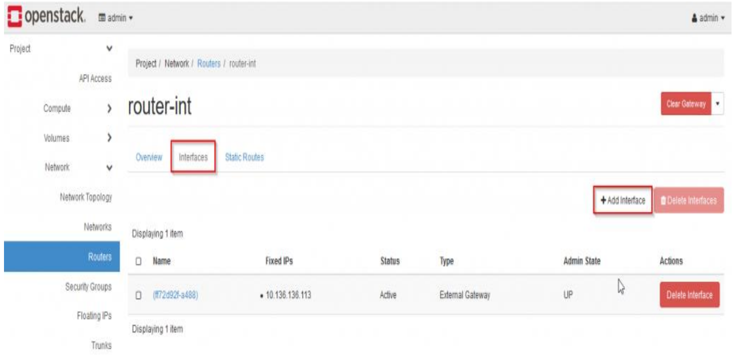


Create Router

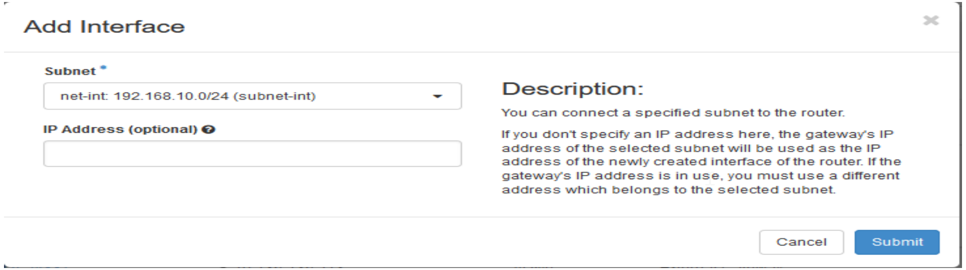
Detail informasi dari router



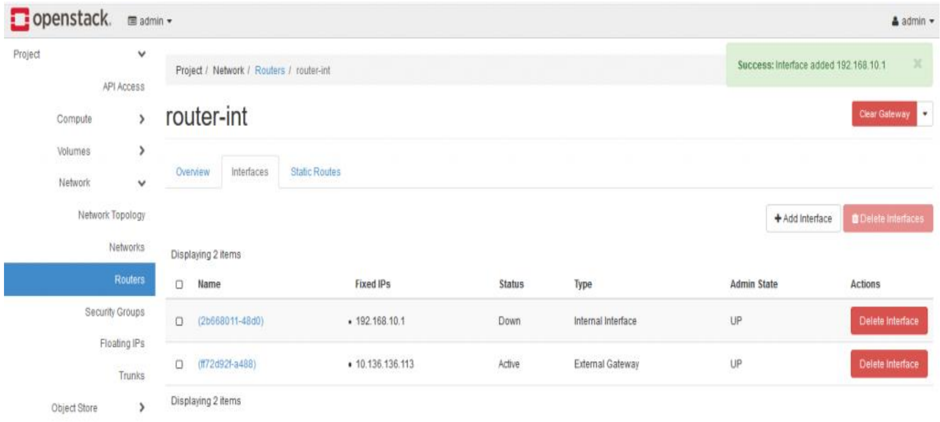
Selanjutnya pindah ke menu **Interface >> Add Interface** untuk menambahkan interface network yang sudah kita buat sebelumnya



Pilih network internal yang sudah dibuat sebelumnya lalu **Submit**



Saat ini interface sudah berhasil di tambahkan

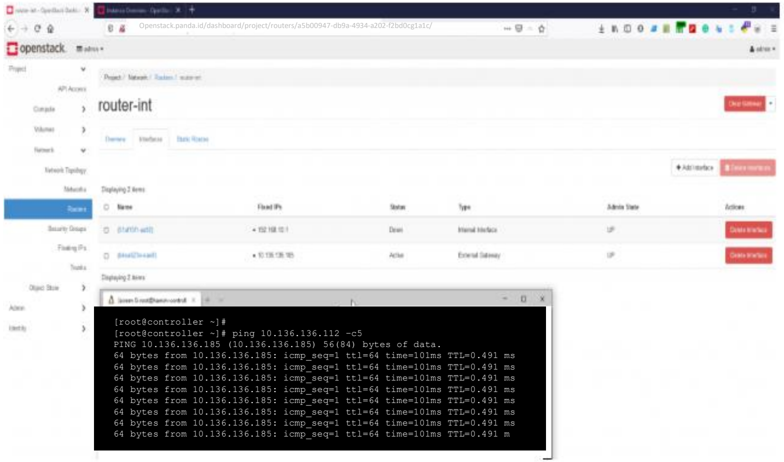


Create Router

Proses pembuatan network internal

Selajutnya test ping dari controller ke IP Gateway atau IP Router yang sudah kita buat sebelumnya.

Openstack.panda.id/dashboard/project/routers/a5b00947-db9a-4934-a202-f2bd0cg1a1c/




```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c5
PING 10.136.136.185 (10.136.136.185) 56(84) bytes of data.
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
```

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ **Openstack: Menambahkan SSH Key**
- ✓ Openstack: Menambah dan Menggunakan Security Group

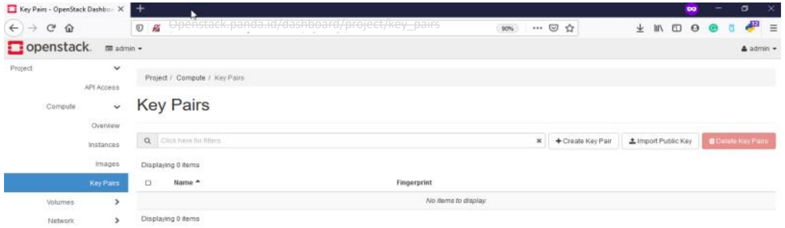
✓ Menambahkan SSH Key

Create SSH keys

Create SSH Keys

Menambahkan SSH Key di openstack. Di openstack untuk menambahkan SSH Key dapat melalui CLI ataupun Horizon, disini kami contohkan menggunakan Horizon.

Login ke *Horizon >> Compute >> Key Pairs*
Openstack.panda.id/dashboard/project/key_pairs



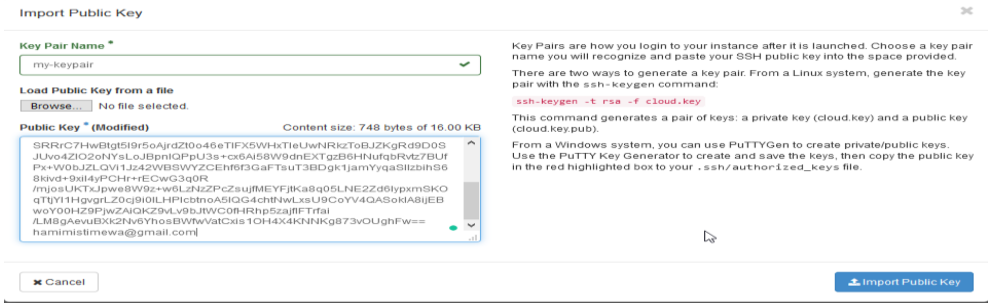
Ada 2 pemilihan terkait SSH Key diantaranya:

a) Import Public Key: Jika Anda sudah mempunyai public key pribadi Anda dapat upload public key nya tanpa create key baru.
b) Create Key Pair: Anda dapat membuat keypair baru dan keypair ini akan Anda gunakan untuk login ke instance

Berikut kami berikan contoh masing – masing pilihan diatas: Import Public Key.

✓ *Import Public Key*

Silakan klik menu **Import Public Key** lalu berikan nama key Anda dan isi public key Anda contoh nya seperti gambar dibawah lalu klik Import Public Key

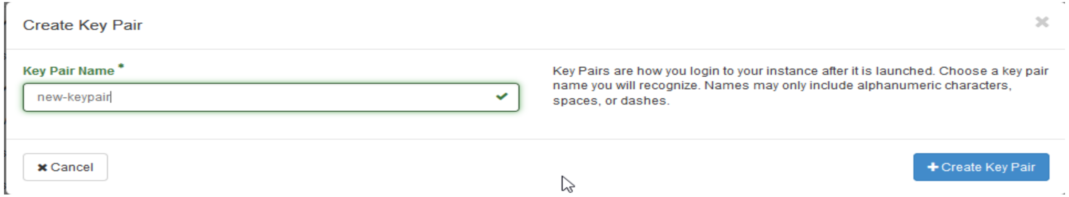


Lab: Kolla Ansible All in One Openstack (Cons’t)

Create SSH Keys

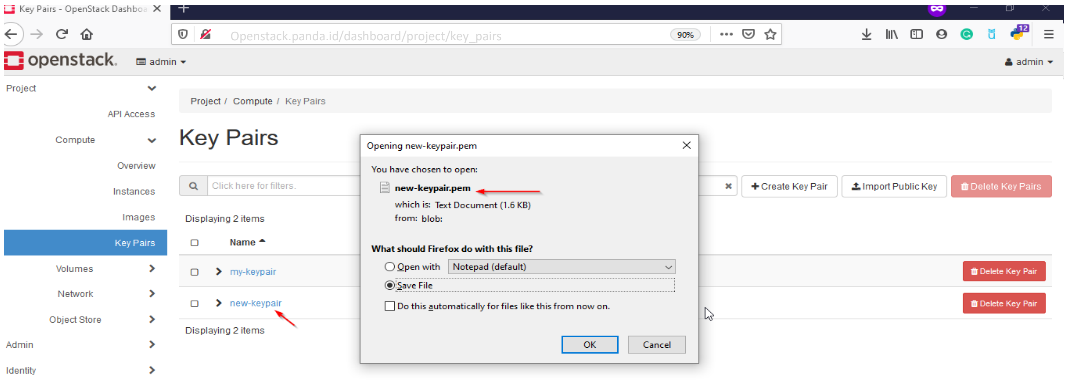
Create Key Pair

Silakan Klik pada **Create Key Pair** lalu isi nama keypair



silakan klik Create Key Pair, setelah nya akan muncul pop seperti Anda, silakan download key pair nya.

Openstack.panda.id/dashboard/project/key_pairs



- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ **Openstack: Menambah dan Menggunakan Security Group**

✓ Menambah Security

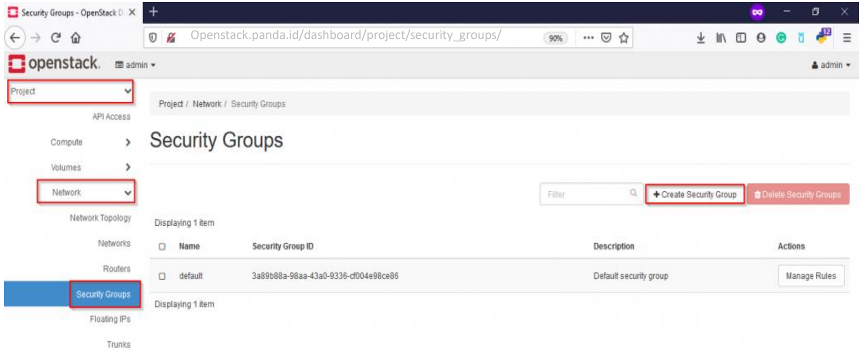
Create Security Group

Create Security Group

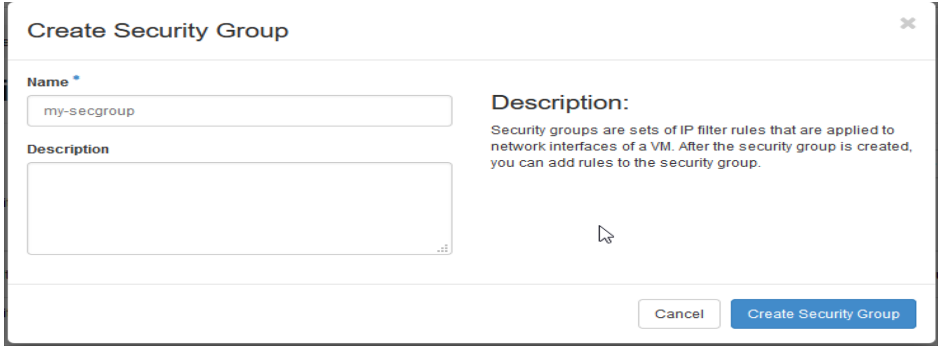
Membuat dan menggunakan security group dapat melalui horizon atau via CLI. Disini kami akan menggunakan horizon untuk menambahkan security group, berikut tahapannya:

Login Horizon >> Project >> Network >> Security Group >> +Create Security Group

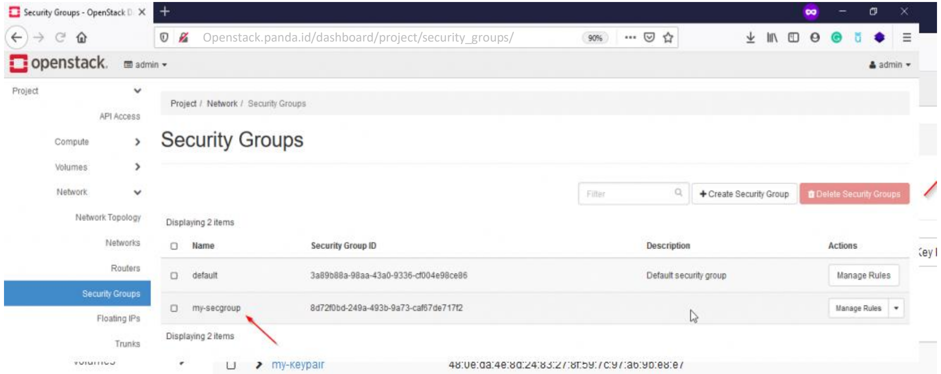
Openstack.panda.id/dashboard/project/security_groups/



Isi nama security group Anda lalu klik Create Security Group



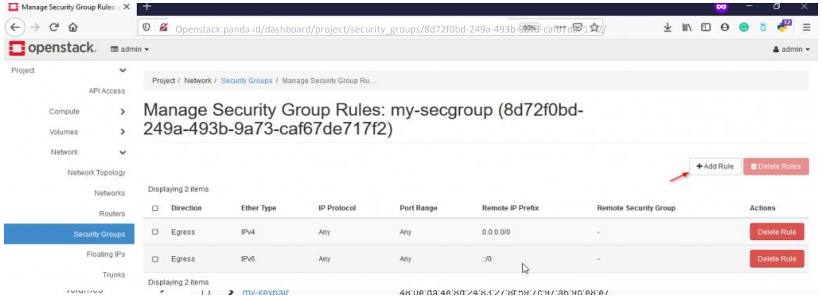
Saat ini security grouup sudah berhasil



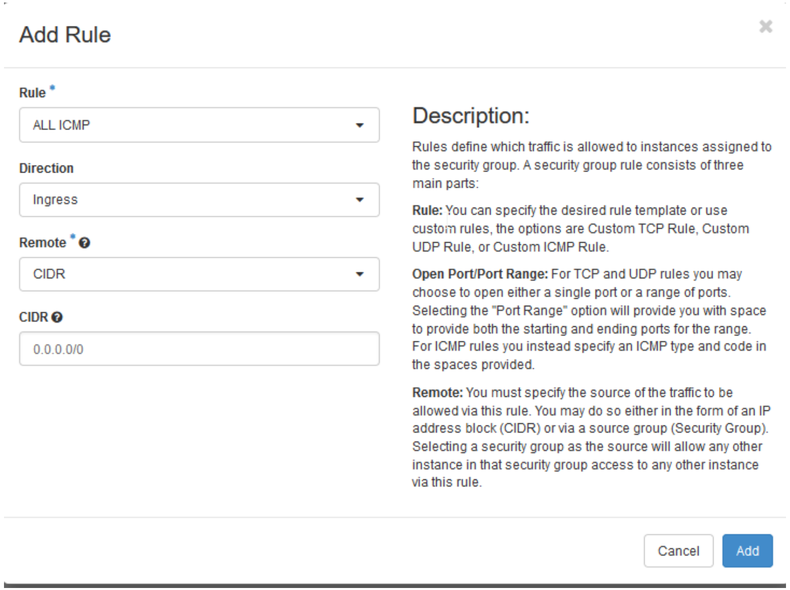
Create Security Group

Terdapat 2 security group (default) dan security group yang sudah dibuat sebelumnya.
Untuk yang default sebenarnya dapat Anda gunakan juga.
Untuk menambahkan port atau menggunakan security group silakan klik menu **Manage Rules** pada security group Anda

Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/

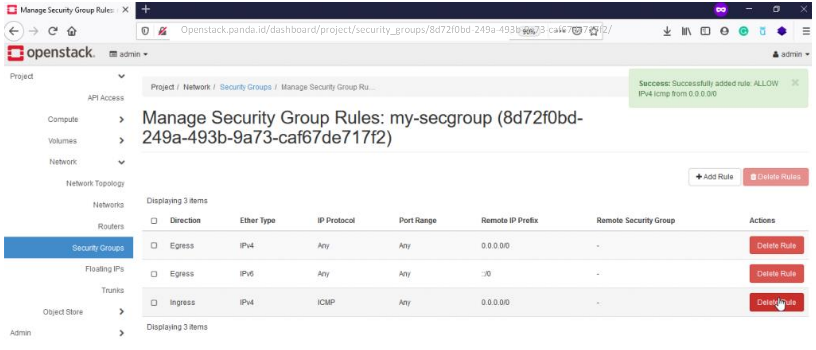


Klik + Add Rule untuk menambahkan rule atau port yang ingin Anda gunakan, misalnya disini kami ingin allow port atau protokol IMCP, lalu klik Add



Create SSH Keys

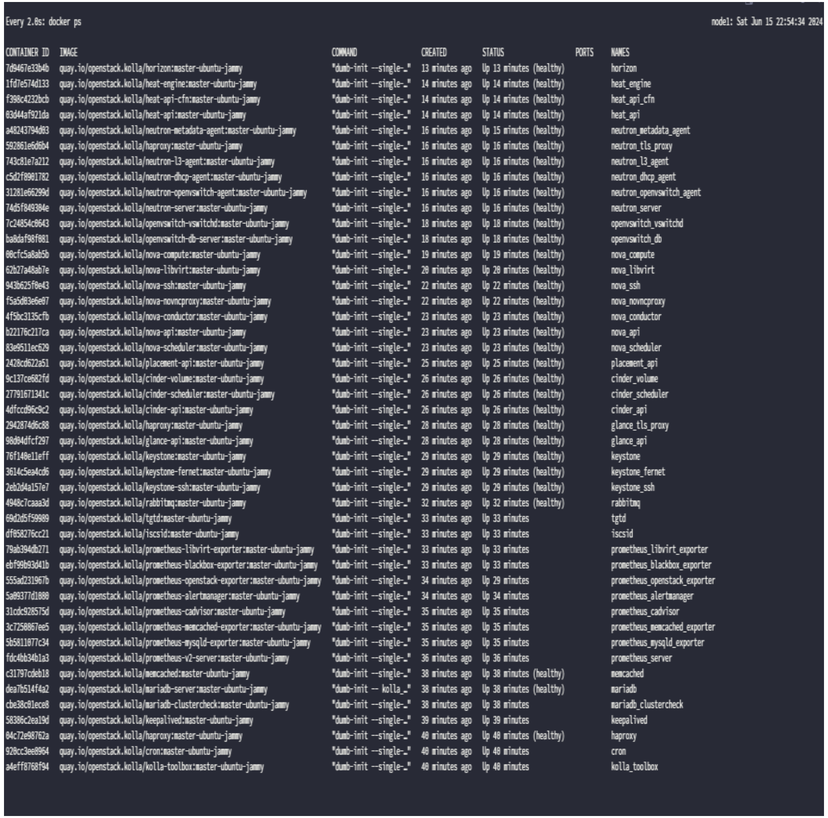
Saat ini rule ICMP sudah dibuat
Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/



Dapat menambahkan rule atau port – port yang Anda inginkan misalnya port ssh, imap, http dan yang lainnya melalui security group

Lab: Kolla Ansible All in One Openstack (Cons't)

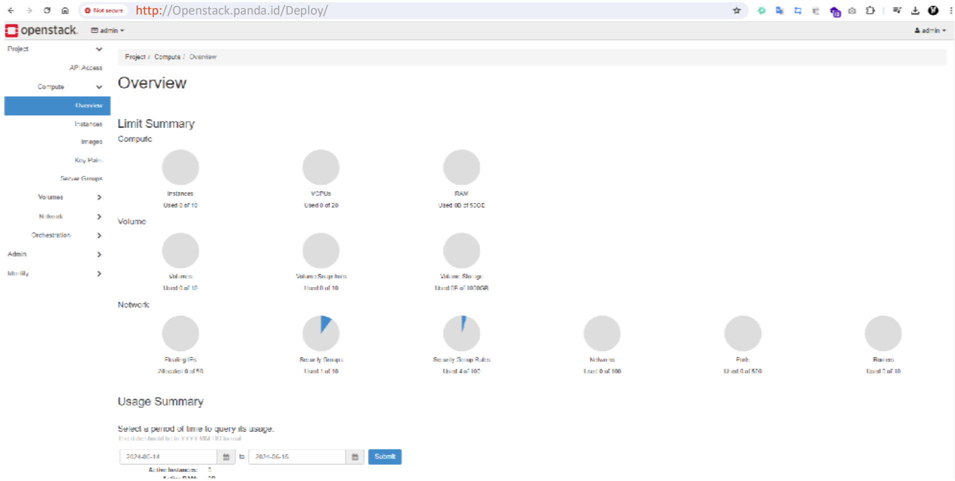
All Service running on Docker



```
Every 2.0s: docker ps                                                                                              node1: Sat Jun 15 22:54:34 2024

CONTAINER ID   IMAGE                                                              COMMAND                  CREATED          STATUS                    PORTS     NAMES
7d9467e33b4b   quay.io/openstack.kolla/horizon:master-ubuntu-jammy                "dumb-init --single-…"   13 minutes ago   Up 13 minutes (healthy)             horizon
1fd7e574d133   quay.io/openstack.kolla/heat-engine:master-ubuntu-jammy            "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_engine
f398c4232bcb   quay.io/openstack.kolla/heat-api-cfn:master-ubuntu-jammy           "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_api_cfn
03d44af921da   quay.io/openstack.kolla/heat-api:master-ubuntu-jammy               "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_api
a48243794d03   quay.io/openstack.kolla/neutron-metadata-agent:master-ubuntu-jammy "dumb-init --single-…"   16 minutes ago   Up 15 minutes (healthy)             neutron_metadata_agent
592861e6d6b4   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy                "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_tls_proxy
743c81e7a212   quay.io/openstack.kolla/neutron-l3-agent:master-ubuntu-jammy       "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_l3_agent
c5d2f8901782   quay.io/openstack.kolla/neutron-dhcp-agent:master-ubuntu-jammy     "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_dhcp_agent
31281e66299d   quay.io/openstack.kolla/neutron-openvswitch-agent:master-ubuntu-jammy "dumb-init --single-…" 16 minutes ago  Up 16 minutes (healthy)             neutron_openvswitch_agent
74d5f849304e   quay.io/openstack.kolla/neutron-server:master-ubuntu-jammy         "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_server
7c24854c0643   quay.io/openstack.kolla/openvswitch-vswitchd:master-ubuntu-jammy   "dumb-init --single-…"   18 minutes ago   Up 18 minutes (healthy)             openvswitch_vswitchd
ba8daf98f081   quay.io/openstack.kolla/openvswitch-db-server:master-ubuntu-jammy  "dumb-init --single-…"   18 minutes ago   Up 18 minutes (healthy)             openvswitch_db
00cfc5a8ab5b   quay.io/openstack.kolla/nova-compute:master-ubuntu-jammy           "dumb-init --single-…"   19 minutes ago   Up 19 minutes (healthy)             nova_compute
62b27a48ab7e   quay.io/openstack.kolla/nova-libvirt:master-ubuntu-jammy           "dumb-init --single-…"   20 minutes ago   Up 20 minutes (healthy)             nova_libvirt
943b625f0e43   quay.io/openstack.kolla/nova-ssh:master-ubuntu-jammy               "dumb-init --single-…"   22 minutes ago   Up 22 minutes (healthy)             nova_ssh
f5a5d03e6e07   quay.io/openstack.kolla/nova-novncproxy:master-ubuntu-jammy        "dumb-init --single-…"   22 minutes ago   Up 22 minutes (healthy)             nova_novncproxy
4f5bc3135cfb   quay.io/openstack.kolla/nova-conductor:master-ubuntu-jammy         "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_conductor
b22176c217ca   quay.io/openstack.kolla/nova-api:master-ubuntu-jammy               "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_api
83e9511ec629   quay.io/openstack.kolla/nova-scheduler:master-ubuntu-jammy         "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_scheduler
2428cd622a51   quay.io/openstack.kolla/placement-api:master-ubuntu-jammy          "dumb-init --single-…"   25 minutes ago   Up 25 minutes (healthy)             placement_api
9c137ce682fd   quay.io/openstack.kolla/cinder-volume:master-ubuntu-jammy          "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_volume
27791671341c   quay.io/openstack.kolla/cinder-scheduler:master-ubuntu-jammy       "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_scheduler
4dfccd96c9c2   quay.io/openstack.kolla/cinder-api:master-ubuntu-jammy             "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_api
2942874d6c88   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy                "dumb-init --single-…"   28 minutes ago   Up 28 minutes (healthy)             glance_tls_proxy
98d04dfcf297   quay.io/openstack.kolla/glance-api:master-ubuntu-jammy             "dumb-init --single-…"   28 minutes ago   Up 28 minutes (healthy)             glance_api
76f140e11eff   quay.io/openstack.kolla/keystone:master-ubuntu-jammy               "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone
3614c5ea4cd6   quay.io/openstack.kolla/keystone-fernet:master-ubuntu-jammy        "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone_fernet
2eb2d4a157e7   quay.io/openstack.kolla/keystone-ssh:master-ubuntu-jammy           "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone_ssh
4948c7caaa3d   quay.io/openstack.kolla/rabbitmq:master-ubuntu-jammy               "dumb-init --single-…"   32 minutes ago   Up 32 minutes (healthy)             rabbitmq
69d2d5f59989   quay.io/openstack.kolla/tgtd:master-ubuntu-jammy                   "dumb-init --single-…"   33 minutes ago   Up 33 minutes                       tgtd
df058276cc21   quay.io/openstack.kolla/iscsid:master-ubuntu-jammy                 "dumb-init --single-…"   33 minutes ago   Up 33 minutes                       iscsid
79ab394db271   quay.io/openstack.kolla/prometheus-libvirt-exporter:master-ubuntu-jammy "dumb-init --single-…" 33 minutes ago Up 33 minutes                       prometheus_libvirt_exporter
ebf99b93d41b   quay.io/openstack.kolla/prometheus-blackbox-exporter:master-ubuntu-jammy "dumb-init --single-…" 33 minutes ago Up 33 minutes                      prometheus_blackbox_exporter
555ad231967b   quay.io/openstack.kolla/prometheus-openstack-exporter:master-ubuntu-jammy "dumb-init --single-…" 34 minutes ago Up 29 minutes                     prometheus_openstack_exporter
5a09377d1880   quay.io/openstack.kolla/prometheus-alertmanager:master-ubuntu-jammy "dumb-init --single-…"  34 minutes ago   Up 34 minutes                       prometheus_alertmanager
31cdc928575d   quay.io/openstack.kolla/prometheus-cadvisor:master-ubuntu-jammy    "dumb-init --single-…"   35 minutes ago   Up 35 minutes                       prometheus_cadvisor
3c7258867ee5   quay.io/openstack.kolla/prometheus-memcached-exporter:master-ubuntu-jammy "dumb-init --single-…" 35 minutes ago Up 35 minutes                     prometheus_memcached_exporter
5b5811077c34   quay.io/openstack.kolla/prometheus-mysqld-exporter:master-ubuntu-jammy "dumb-init --single-…" 35 minutes ago Up 35 minutes                        prometheus_mysqld_exporter
fdc4bb34b1a3   quay.io/openstack.kolla/prometheus-v2-server:master-ubuntu-jammy   "dumb-init --single-…"   36 minutes ago   Up 36 minutes                       prometheus_server
c31797cdeb18   quay.io/openstack.kolla/memcached:master-ubuntu-jammy              "dumb-init --single-…"   38 minutes ago   Up 38 minutes (healthy)             memcached
dea7b514f4a2   quay.io/openstack.kolla/mariadb-server:master-ubuntu-jammy         "dumb-init -- kolla_…"   38 minutes ago   Up 38 minutes (healthy)             mariadb
cbe38c01ece8   quay.io/openstack.kolla/mariadb-clustercheck:master-ubuntu-jammy   "dumb-init --single-…"   38 minutes ago   Up 38 minutes                       mariadb_clustercheck
58386c2ea19d   quay.io/openstack.kolla/keepalived:master-ubuntu-jammy             "dumb-init --single-…"   39 minutes ago   Up 39 minutes                       keepalived
04c72e98762a   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy                "dumb-init --single-…"   40 minutes ago   Up 40 minutes (healthy)             haproxy
920cc3ee8964   quay.io/openstack.kolla/cron:master-ubuntu-jammy                   "dumb-init --single-…"   40 minutes ago   Up 40 minutes                       cron
a4eff8768f94   quay.io/openstack.kolla/kolla-toolbox:master-ubuntu-jammy          "dumb-init --single-…"   40 minutes ago   Up 40 minutes                       kolla_toolbox
```

Horizon



Lab: Kolla Ansible All in One Openstack (Cons’t)

Network Agent

```
root@node1:~# openstack network agent list
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host  | Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
| 53099b35-21c1-4740-acd6-34781fbbc720 | Open vSwitch agent | node1 | None              | :-)   | UP    | neutron-openvswitch-agent |
| 6605786d-3110-43ee-b526-741b2b43abb9 | L3 agent           | node1 | nova              | :-)   | UP    | neutron-l3-agent          |
| 8ef397db-e6c9-418a-a532-dc6ccc330d97 | DHCP agent         | node1 | nova              | :-)   | UP    | neutron-dhcp-agent        |
| b3c9da4e-d28e-4c51-b767-c4c8e4d208da | Metadata agent     | node1 | None              | :-)   | UP    | neutron-metadata-agent    |
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
```

Compute Service

```
root@node1:~# openstack compute service list
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
| ID                                   | Binary         | Host  | Zone     | Status  | State | Updated At                 |
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
| 058a6902-5f6c-4907-ad8e-84ec76c75f69 | nova-scheduler | node1 | internal | enabled | up    | 2024-06-15T15:57:17.000000 |
| 7c2550c7-799b-495e-ac1a-71bcab069b41 | nova-conductor | node1 | internal | enabled | up    | 2024-06-15T15:57:13.000000 |
| b4998b8f-c07a-4b0e-9330-df2b1a9b0206 | nova-compute   | node1 | nova     | enabled | up    | 2024-06-15T15:57:11.000000 |
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
root@node1:~#
```

Volume Service

```
root@node1:~# openstack volume service list
+------------------+-------------+------+---------+-------+----------------------------+
| Binary           | Host        | Zone | Status  | State | Updated At                 |
+------------------+-------------+------+---------+-------+----------------------------+
| cinder-scheduler | node1       | nova | enabled | up    | 2024-06-15T15:59:53.000000 |
| cinder-volume    | node1@lvm-1 | nova | enabled | up    | 2024-06-15T15:59:44.000000 |
+------------------+-------------+------+---------+-------+----------------------------+
root@node1:~#
```

Compute Service

Openstack.panda.id/dashboard/admin/images

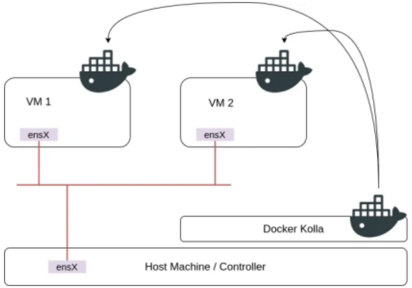
Openstack.panda.id/Deploy/

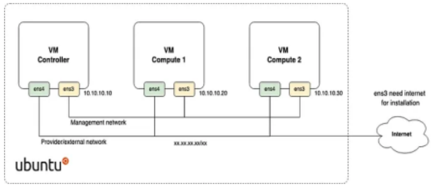
Deploy All-in-One OpenStack with Kolla-Ansible on Ubuntu 22.04

OpenStack Series #2 — Install openstack menggunakan Kolla Ansible (Multi-nodes)

Prasyarat

1. Bare-metal / Virtual Machine (Minimal 1 node untuk metode All In One dan minimal 2 node untuk metode Multinode)
2. Minimal 2 NIC per VM dengan 1 interface yang tidak diberikan ip
3. Disk tambahan untuk backend storage (At Least 50gb)
4. Ram minimal 8gb
5. Koneksi ke Internet
6. Mempunyai akses root maupun user sudoer






```
$ openstack service list
+----------------------------------+-------------+----------------+
| ID                               | Name        | Type           |
+----------------------------------+-------------+----------------+
| 10cb2de52c404662aa6ca3ccdb8f2cf7 | cinderv2    | volumev2       |
| 250a6a7cf4d44f0c9cdd2a39f8203ccc | cinderv3    | volumev3       |
| 2d99d37fc498465591faa9f44580d991 | placement   | placement      |
| 4001de18bf5c45f5a1467a31ea5a2afd | heat-cfn    | cloudformation |
| 505333b0b90b4962a37d55c44bc940f9 | keystone    | identity       |
| 612c645a37e848d49af9f6f53dc6e8ac | nova_legacy | compute_legacy |
| 87c2edd540c84c8296251ea284a14398 | neutron     | network        |
| 94e54356818d43af8ed7f4e1a6da28f1 | nova        | compute        |
| a9f6eb791ec740c183ecff67dc7c7b08 | glance      | image          |
| eaf1f96955244acfbf5124e42c32f4e6 | heat        | orchestration  |
+----------------------------------+-------------+----------------+
```

Openstack.panda.id/dashboard/admin/images

Openstack.panda.id/Deploy/

Openstack 2024.2 with Kolla Ansible

Openstack 2024.2 with Kolla Ansible (Cons't)

Node Hostname	*Node Role*	*vCPU*	*Memory*	*RootDisk*	*ManagementNet*	*StorageNet*
Deploy01adm01	Ansible	2 Core	2GB	20GB	10.78.78.199	-
Deploy01con01	Controller	16 Core	16GB	40GB	10.78.78.201	10.79.79.201
Deploy01con02	Controller	16 Core	16GB	40GB	10.78.78.202	10.79.79.202
Deploy01con03	Controller	16 Core	16GB	40GB	10.78.78.203	10.79.79.203
Deploy01hpv01	Hypervisor	32 Core	32GB	40GB	10.78.78.204	10.79.79.204
Deploy01hpv02	Hypervisor	32 Core	32GB	40GB	10.78.78.205	10.79.79.205

Openstack 2024.2 with Kolla Ansible (Cons't)

Interface mapping:

- ✓ eth0 for Public Network
- ✓ eth1 for Management Network
- ✓ eth2 for Storage Network with Jumbo frames
- ✓ eth4 for Tunnel/Self Service/Guest Network with Jumbo frames

We can review network configuration in **sample-netplan.yaml**


```yaml
network:
 version: 2
 ethernets:
  eth0:
   match:
    macaddress: "BC:24:11:D3:02:2F"
   set-name: "eth0"
   critical: true
  eth1:
   match:
    macaddress: "BC:24:11:77:6F:A2"
   set-name: "eth1"
   addresses:
   - "10.78.78.201/24"
   nameservers:
    addresses:
    - 10.78.78.103
    - 10.78.78.104
    search:
    - Deploy.is-a.dev
   routes:
   - to: "default"
    via: "10.78.78.254"
   critical: true
  eth2:
   match:
    macaddress: "BC:24:11:D6:06:52"
   set-name: "eth2"
   addresses:
   - "10.79.79.201/24"
   critical: true
   mtu: 9000
  eth3:
   match:
    macaddress: "BC:24:11:EF:59:FD"
   set-name: "eth3"
   addresses:
   - "10.79.80.201/24"
   critical: true
   mtu: 9000
```

Openstack 2024.2 with Kolla Ansible (Cons't)

Installing NFS Server for Glance Service in Deploy01adm01 node

```
sudo apt update
sudo apt install nfs-kernel-server
sudo mkdir -p /data
sudo chown nobody:nogroup /data
echo "/data *(rw,sync,no_subtree_check)" >> /etc/exports
sudo systemctl restart nfs-kernel-server
sudo exportfs -v
```

Execute in all nodes

Mapping static hostname in /etc/hosts

```
cat <<EOF | tee -a /etc/hosts
10.78.78.199 Deploy01adm01 Deploy01adm01.Deploy.is-a.dev
10.78.78.201 Deploy01con01 Deploy01con01.Deploy.is-a.dev
10.78.78.202 Deploy01con02 Deploy01con02.Deploy.is-a.dev
10.78.78.203 Deploy01con02 Deploy01con03.Deploy.is-a.dev
10.78.78.204 Deploy01hpv01 Deploy01hpv01.Deploy.is-a.dev
10.78.78.205 Deploy01hpv02 Deploy01hpv02.Deploy.is-a.dev

10.78.78.200 os-int
10.78.78.100 os-ext
EOF
```

Install packages and dependencies

```
sudo apt install -y git gcc libssl-dev libffi-dev \
 python3-venv python3-dev python3-selinux python3-setuptools \
 python3-pip python3-docker nfs-common
```

Mount nfs volume for all node

```
sudo mkdir /data cat <<EOF | sudo tee -a /etc/fstab 10.78.78.199:/data /data nfs defaults,_netdev 0 0
EOF
 sudo mount /data
```

Create virtual environment for kolla-ansible

```
python3 -m venv kolla-venv
cd kolla-venv
source bin/activate
 pwd
```

```
panda@Deploy01adm01: ~ cd kolla-venv
panda@Deploy01adm01: ~ /kolla-venv$ source bin/activate
(kola-venv) panda@Deploy01adm01: ~ /kolla-venv $ pwd
/home/panda/ kolla-venv
(kola-venv) panda@Deploy01adm01: ~ /kolla-venv $
```

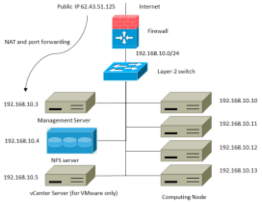
Update pip and install ansible

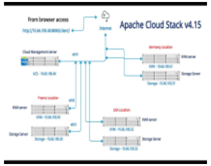
```
pip install -U pip pip install 'ansible>=8,<9'
```

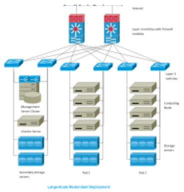
All in one Multi Node cloud infrastructure (compute, Network, Storage)

Apache Cloudstack

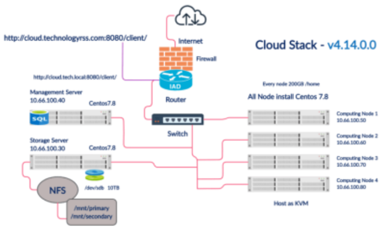
Small-Scale Deployment



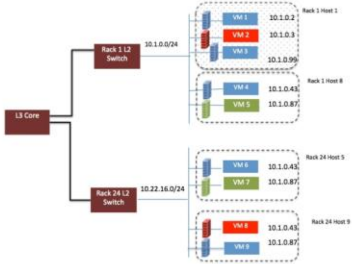


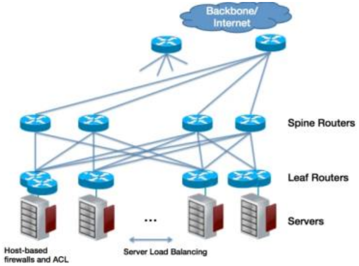


Cloud Stack



CloudStack Basic Networking





Deploy OpenStack using Kolla-Ansible (Cons't)

Lab Environment

Hosts

```
Hostname              | Role and Spec.
------------------------------------------------------------------------------------------------
GPU10                 | deploy (Ubuntu 22.04)
 GPU111               | controller (Ubuntu 20.04, 4CPU, 32G RAM, 192.168.1.23)
GPU71                 | compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.24)
GPU72                 | compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.25)
GPU73                 | compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.26)
```

Hosts

```
eno1: internal (192.168.1.0/24)
eno2: External
```

NOTE: The eno2 interface is a "USB" NIC, rename it to eno2 base on mac address on each host.
Account username: "ubuntu"

Prerequest

Add hosts to /etc/hosts in all hosts

```
amel@GPU10:~$ ssh-keygen
amel@GPU10:~$ ssh-copy-id GPU111
amel@GPU10:~$ ssh-copy-id GPU71
amel@GPU10:~$ ssh-copy-id GPU72
amel@GPU10:~$ ssh-copy-id GPU73
```

GPU == GPU

Configure host network interface. For example on GPU111

```
ubuntu@GPU111:~$ cat /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    eno1:
      dhcp4: false
      link-local: []
      addresses:
        - 192.168.1.23/24
      gateway4: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
    eno2:
      dhcp4: false
      link-local: []
      match:
        macaddress: f8:e4:3b:87:00:1f
      set-name: eno2
  version: 2
```

```
ubuntu@GPU111:~$ sudo netplan try
ubuntu@GPU111:~$ sudo netplan apply
```

NOTE: Do not configure IP address on eno2 interface.

Add user to /etc/sudoers for no password sudo

```
ubuntu ALL=(ALL) NOPASSWD:ALL
```

NOTE: Refer "Use local docker registry to deploy" section for using local registry

Deploy OpenStack using Kolla-Ansible (Cons’t)

1. Setup kolla ansible on deploy node

Install dependency packages on

```
amel@GPU10:~$ sudo apt update -y
amel@GPU10:~$ sudo apt install python3-dev libffi-dev gcc libssl-dev -y
```

Install python virtual environment for kolla-ansible

```
amel@GPU10:~$ sudo apt install python3-venv –y
```

```
amel@GPU10:~$ python3 -m venv ~/Data/kolla-ansible-venv
```

```
amel@GPU10:~$ source ~/Data/kolla-ansible-venv/bin/activate (kolla-ansible-venv) amel@GPU10:~$
```

```
(kolla-ansible-venv) amel@GPU10:~$ pip install -U pip
```

```
(kolla-ansible-venv) amel@GPU10:~$ pip install 'ansible>=4,<6‘
```

Install Kolla-ansible

```
(kolla-ansible-venv) amel@GPU10:~$ pip install git+https://opendev.org/openstack/kolla-ansible@master
```

```
(kolla-ansible-venv) amel@GPU10:~$ sudo mkdir -p /etc/
kolla(kolla-ansible-venv) amel@GPU10:~$ sudo chown $USER:$USER /etc/kolla/
(kolla-ansible-venv) amel@GPU10:~$ ls -ld /etc/kolla/
drwxr-xr-x 2 amel amel 4096 Nov  5 16:01 /etc/kolla/
```

```
(kolla-ansible-venv) amel@GPU10:~$ cp -r ~/Data/kolla-ansible-venv/share/kolla-ansible/etc_examples//kolla/* /etc/kolla/
(kolla-ansible-venv) amel@GPU10:~$ cp ~/Data/kolla-ansible-venv/share/kolla-ansible/ansible/inventory/multinode /etc/kolla/
```

NOTE: for all-in-one deploy, use ~/Data/kolla-ansible-venv/share/kolla-ansible/ansible/inventory/all-in-one

Install Ansible Galaxy requirements

```
(kolla-ansible-venv) amel@GPU10:~$ kolla-ansible install-deps
Installing Ansible Galaxy dependencies
Starting galaxy collection install process
Process install dependency map
Cloning into '/home/amel/.ansible/tmp/ansible-local-111805_n2pn0om/tmp5syatw1g/ansible-collection-kolla9pniyzg2'...
remote: Enumerating objects: 150, done.
remote: Counting objects: 100% (150/150), done.
remote: Compressing objects: 100% (97/97), done.
remote: Total 409 (delta 120), reused 53 (delta 53), pack-reused 259
Receiving objects: 100% (409/409), 79.20 KiB | 477.00 KiB/s, done.
Resolving deltas: 100% (159/159), done.
Already on 'master'
Your branch is up to date with 'origin/master'.
Starting collection install process
Installing 'openstack.kolla:1.0.0' to '/home/amel/.ansible/collections/ansible_collections/openstack/kolla'
Created collection for openstack.kolla:1.0.0 at /home/amel/.ansible/collections/ansible_collections/openstack/kolla
openstack.kolla:1.0.0 was installed successfully
(kolla-ansible-venv) amel@GPU10:~$
```

Deploy OpenStack using Kolla-Ansible (Cons’t)

Configure Ansible

```
(kolla-ansible-venv) amel@GPU10:~$ cat /etc/ansible/ansible.cfg
[default]
deprecation_warnings=False
host_key_checking=False
pipelining=True
forks=100
```

Prepare initial configuration

```
(kolla-ansible-venv) amel@GPU10:~$ cd /etc/kolla/
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ls -l
total 52
-rw-rw-r-- 1 amel amel 32632 Nov  5 16:07 globals.yml
-rw-rw-r-- 1 amel amel  9490 Nov  5 16:25 multinode
-rw-rw-r-- 1 amel amel  5163 Nov  5 16:07 passwords.yml
```

Edit multinode inventory file

```
# These initial groups are the only groups required to be modified. The
# additional groups are for more control of the environment.
[control]
# These hostname must be resolvable from your deployment host
GPU111

# The above can also be specified as follows:
#control[01:03]     ansible_user=kolla

# The network nodes are where your l3-agent and loadbalancers will run
# This can be the same as a host in the control group
[network:children]
control

[compute]
GPU7[1:3]

[monitoring:children]
control

# When compute nodes and control nodes use different interfaces,
# you need to comment out "api_interface" and other interfaces from the globals.yml
# and specify like below:
#compute01 neutron_external_interface=eth0 api_interface=em1 tunnel_interface=em1

[storage:children]
control
```

NOTE: I use "network:children", "storage:children" and "monitoring:children" to add "control" as children group. Use hostname if you do not want to use children groups.

Deploy OpenStack using Kolla-Ansible (Cons't)

Check inventory

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ansible -i multinode all -m ping
[WARNING]: Invalid characters were found in group names but not replaced,
use -vvvv to see details
localhost | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
GPU111 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
GPU72 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
GPU73 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
GPU71 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Set Kolla passwords

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-genpwd
```

Edit global.yml file as below

```
48 # This should be a VIP, an unused IP on your network that will float between
49 # the hosts running keepalived for high-availability. If you want to run an
50 # All-In-One without haproxy and keepalived, you can set enable_haproxy to
no
51 # in "OpenStack options" section, and set this value to the IP of your
52 # 'network_interface' as set in the Networking section below.
53 #kolla_internal_vip_address: "10.10.10.254"
54 kolla_internal_vip_address: "192.168.1.250"

111 ###############################
112 # Neutron - Networking Options
113 ###############################
114 # This interface is what all your api services will be bound to by default.
115 # Additionally, all vxlan/tunnel and storage network traffic will go over this
116 # interface by default. This interface must contain an IP address.
117 # It is possible for hosts to have non-matching names of interfaces - these
can
118 # be set in an inventory file per host or per group or stored separately, see
119 #     http://docs.ansible.com/ansible/intro_inventory.html
120 # Yet another way to workaround the naming problem is to create a bond
for the
121 # interface on all hosts and give the bond name here. Similar strategy can be
122 # followed for other types of interfaces.
123 #network_interface: "eth0"
124 network_interface: "eno1"

149 # This is the raw interface given to neutron as its external network port. Even
150 # though an IP address can exist on this interface, it will be unusable in most
151 # configurations. It is recommended this interface not be configured with
any IP
152 # addresses for that reason.
153 #neutron_external_interface: "eth1"
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Create "nfs_shares" file with nfs configuration

```
amel@GPU10:/etc/kolla$ mkdir config
amel@GPU10:/etc/kolla$ cat config/nfs_shares
GPU111:/kolla_nfs
```

2. Setup NFS server on controller for cinder backend

```
root@GPU111:~# apt install nfs-kernel-server -y
root@GPU111:~# cat /etc/exports
/kolla_nfs 192.168.1.0/24(rw,sync,no_root_squash)
root@GPU111:~# mkdir /kolla_nfs
root@GPU111:~# systemctl start nfs-kernel-server
```

```
root@GPU111:~# systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/lib/systemd/system/nfs-server.service; enabled; vendor
preset: enabled)
  Drop-In: /run/systemd/generator/nfs-server.service.d
           └─order-with-mounts.conf
   Active: active (exited) since Sat 2025-18-05 11:51:21 UTC; 24min ago
  Process: 67275 ExecStartPre=/usr/sbin/exportfs -r (code=exited,
status=0/SUCCESS)
  Process: 67276 ExecStart=/usr/sbin/rpc.nfsd $RPCNFSDARGS (code=exited,
status=0/SUCCESS)
 Main PID: 67276 (code=exited, status=0/SUCCESS)

Nov 05 11:51:19 GPU111 systemd[1]: Starting NFS server and services...
Nov 05 11:51:19 GPU111 exportfs[67275]: exportfs: /etc/exports [1]: Neither
'subtree_check' or 'no_subtree_check' specified for export
"192.168.1.0/24:/kolla_nfs".
Nov 05 11:51:19 GPU111 exportfs[67275]:   Assuming default behaviour
('no_subtree_check').
```

3. Deployment

Bootstrap servers with kolla deploy dependencies

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode bootstrap-servers

... output omit

PLAY RECAP ***************************************************************************
localhost      : ok=2   changed=0   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0
GPU111         : ok=31  changed=15  unreachable=0   failed=0   skipped=21  rescued=0   ignored=0
GPU71          : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
GPU72          : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
GPU73          : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
```

NOTE: if bootstrap-servers playbook failed with error. Failed to update apt cache: W:Updating from such a repository can't be done securely, and is therefore disabled by default, re-run the playbook**.**

Do pre-deployment checks for hosts

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode prechecks

... output omit

PLAY RECAP ***************************************************************************
localhost      : ok=11  changed=0   unreachable=0   failed=0   skipped=14  rescued=0   ignored=0
GPU111         : ok=93  changed=0   unreachable=0   failed=0   skipped=159 rescued=0   ignored=0
GPU71          : ok=29  changed=0   unreachable=0   failed=0   skipped=30  rescued=0   ignored=0
GPU72          : ok=28  changed=0   unreachable=0   failed=0   skipped=27  rescued=0   ignored=0
GPU73          : ok=28  changed=0   unreachable=0   failed=0   skipped=27  rescued=0   ignored=0
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Run actual deployment

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode deploy

... output omit

PLAY RECAP ****************************************************************************
localhost     : ok=4    changed=0    unreachable=0   failed=0   skipped=0    rescued=0   ignored=0
GPU111        : ok=287  changed=122  unreachable=0   failed=0   skipped=220  rescued=0   ignored=0
GPU71         : ok=59   changed=33   unreachable=0   failed=0   skipped=45   rescued=0   ignored=0
GPU72         : ok=54   changed=33   unreachable=0   failed=0   skipped=45   rescued=0   ignored=0
GPU73         : ok=54   changed=33   unreachable=0   failed=0   skipped=46   rescued=0   ignored=0
```

4. Access OpenStack

Install openstack client

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ pip install python-openstackclient -c
https://releases.openstack.org/constraints/upper/master
```

Generate admin openrc and clouds.yaml

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ls -l
total 96
-rw------- 1 amel amel   565 Nov  5 23:38 admin-openrc.sh
-rw------- 1 amel amel   573 Nov  5 23:38 clouds.yaml
drwxrwxr-x 2 amel amel  4096 Nov  5 22:54 config
-rw-rw-r-- 1 amel amel 32846 Nov  5 22:27 globals.yml
-rw-rw-r-- 1 amel amel  9475 Nov  5 16:59 multinode
-rw-rw-r-- 1 amel amel 34047 Nov  5 17:10 passwords.yml
```

Access method 1: source admin-openrc.sh

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ source admin-openrc.sh
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ openstack catalog list
+-----------+---------------+-------------------------------------------------------------------------+
| Name      | Type          | Endpoints                                                               |
+-----------+---------------+-------------------------------------------------------------------------+
| nova      | compute       | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:8774/v2.1                                |
|           |               | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:8774/v2.1                              |
|           |               |                                                                         |
| cinderv3  | volumev3      | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2 |
|           |               | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2 |
|           |               |                                                                         |
| placement | placement     | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:8780                                   |
|           |               | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:8780                                     |
|           |               |                                                                         |
| glance    | image         | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:9292                                   |
|           |               | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:9292                                     |
|           |               |                                                                         |
| keystone  | identity      | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:5000                                     |
|           |               | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:5000                                   |
|           |               |                                                                         |
| neutron   | network       | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:9696                                     |
|           |               | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:9696                                   |
|           |               |                                                                         |
| heat-cfn  | cloudformation | RegionOne                                                              |
|           |               |   internal: http://192.168.1.250:8000/v1                                |
|           |               | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:8000/v1                                  |
|           |               |                                                                         |
| heat      | orchestration | RegionOne                                                               |
|           |               |   public: http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2 |
|           |               | RegionOne                                                               |
|           |               |   internal: http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2 |
|           |               |                                                                         |
+-----------+---------------+-------------------------------------------------------------------------+

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ openstack compute service list
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Access method 2: use clouds.yaml

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat clouds.yaml
clouds:
 kolla-admin:
  auth:
   auth_url: http://192.168.1.250:5000
   project_domain_name: Default
   user_domain_name: Default
   project_name: admin
   username: admin
   password: Lo6CMreQQAwHmp3sCjR0NqaqDm0fCoX7I0OBVY1c
  region_name: RegionOne
 kolla-admin-internal:
  auth:
   auth_url: http://192.168.1.250:5000
   project_domain_name: Default
   user_domain_name: Default
   project_name: admin
   username: admin
   password: Lo6CMreQQAwHmp3sCjR0NqaqDm0fCoX7I0OBVY1c
  interface: internal
  region_name: RegionOne
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./clouds.yaml /etc/openstack/
OR
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir ~/.config/openstack
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./clouds.yaml
~/.config/openstack
OR
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ export
OS_CLIENT_CONFIG_FILE=/etc/kolla/clouds.yaml
```

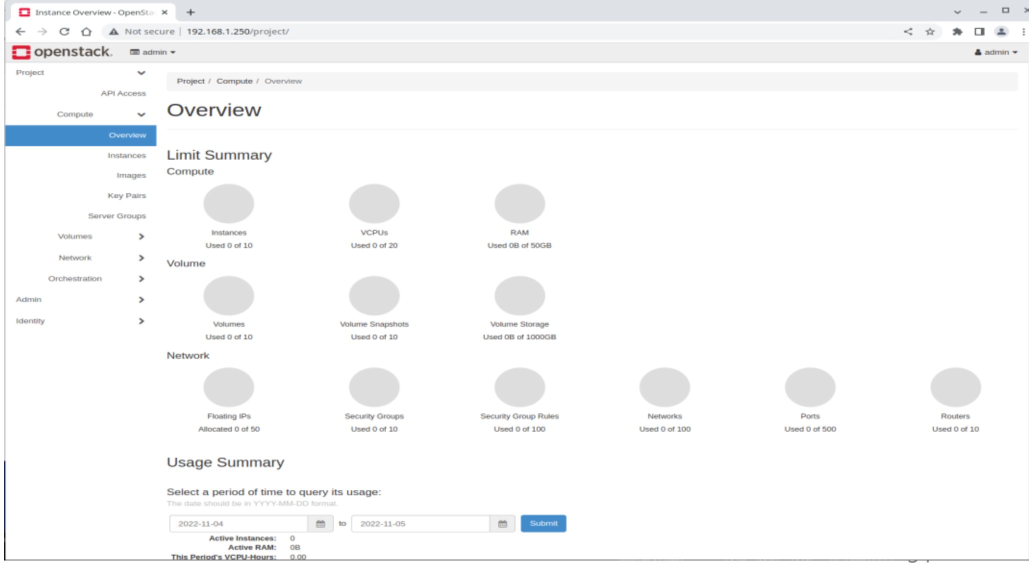
openstack command with "–os-cloud" option to choose cloud

(kolla-ansible-venv) amel@GPU10:~$ openstack --os-cloud kolla-admin compute service list

ID	Binary	Host	Zone	Status	State	Updated At
9cb2e662-0446-417d-9ecb-d0ef5679459e	nova-scheduler	GPU111	internal	enabled	up	2025-18-05T13:05:22.000000
19dc19a4-4d78-4909-95a4-60fda0e62839	nova-conductor	GPU111	internal	enabled	up	2025-18-05T13:05:18.000000
795eefbd-9461-4462-8a68-125e77027a5b	nova-compute	GPU71	nova	enabled	up	2025-18-05T13:05:21.000000
b1734c47-918c-4aee-9fc0-78326b4dde8e	nova-compute	GPU73	nova	enabled	up	2025-18-05T13:05:21.000000
feb93682-0dab-4dc0-95cc-872d4196339d	nova-compute	GPU72	nova	enabled	up	2025-18-05T13:05:21.000000

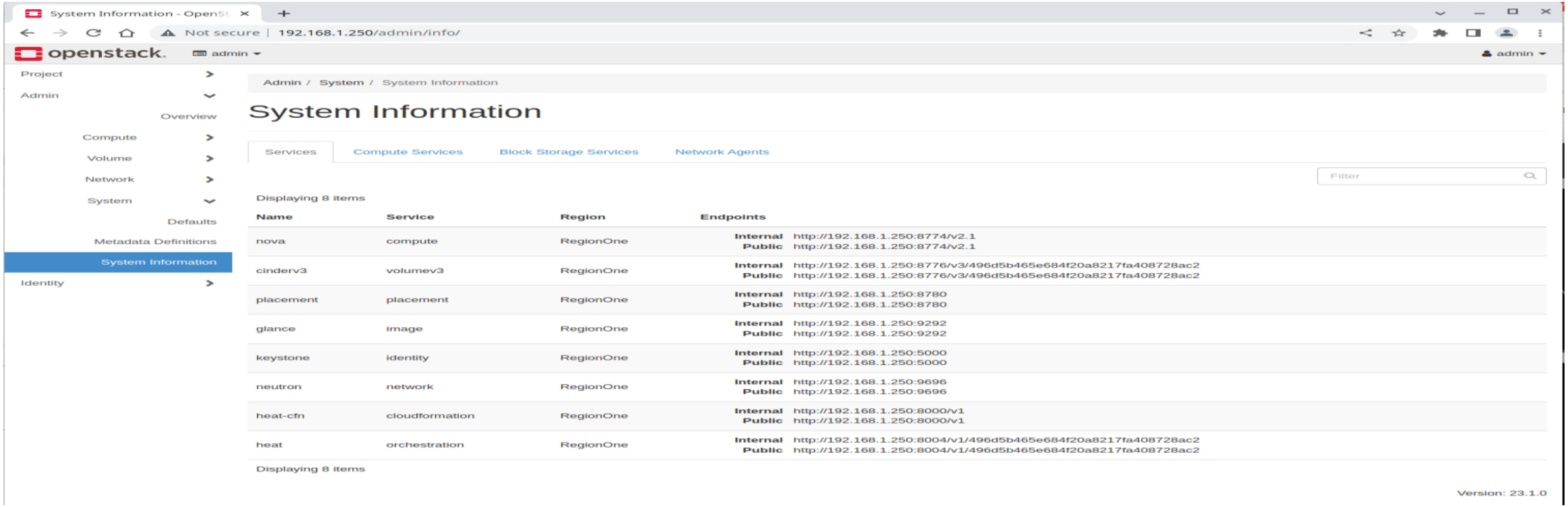
5. Access Web GUI (Horizon)

Get admin password from admin-open.rc or clouds.yaml



Deploy OpenStack using Kolla-Ansible (Cons’t)

System Information



Deploy OpenStack using Kolla-Ansible (Cons't)

Extra: create examples for demo

NOTE: copy "clouds.yaml" in controller node to /etc/kolla/ first.

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ~/Data/kolla-ansible-venv/share/kolla-ansible/init-runonce
Checking for locally available cirros image.
None found, downloading cirros image.
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 20.1M  100 20.1M    0     0  2928k      0  0:00:07  0:00:07 --:--:-- 4371k
Creating glance image.
+------------------+------------------------------------------------------------+
| Field            | Value                                                      |
+------------------+------------------------------------------------------------+
| container_format | bare                                                       |
| created_at       | 2025-18-06T01:03:54Z                                       |
| disk_format      | qcow2                                                      |
| file             | /v2/images/1fa43af3-ac22-4dba-85e9-04b21807c5d4/file       |
| id               | 1fa43af3-ac22-4dba-85e9-04b21807c5d4                       |
| min_disk         | 0                                                          |
| min_ram          | 0                                                          |
| name             | cirros                                                     |
| owner            | 496d5b465e684f20a8217fa408728ac2                           |
| properties       | os_hidden='False', os_type='linux', owner_specified.openstack.md5='', owner_specified.openstack.object='images/cirros', owner_specified.openstack.sha256='' |
| protected        | False                                                      |
| schema           | /v2/schemas/image                                          |
| status           | queued                                                     |
| tags             |                                                            |
| updated_at       | 2025-18-06T01:03:54Z                                       |
| visibility       | public                                                     |
+------------------+------------------------------------------------------------+
Configuring neutron.
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2025-18-06T01:03:56Z                 |
| description             |                                      |
| distributed             | False                                |
| enable_ndp_proxy        | None                                 |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| id                      | d8adf379-d03a-4c58-80cb-9cf7003524a6 |
| name                    | demo-router                          |
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ openstack --os-cloud=kolla-admin server create \
    --image cirros \
    --flavor m1.tiny \
    --key-name mykey \
    --network demo-net \
    demo1
+-------------------------------------+-----------------------------------------------+
| Field                               | Value                                         |
+-------------------------------------+-----------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                        |
| OS-EXT-AZ:availability_zone         |                                               |
| OS-EXT-SRV-ATTR:host                | None                                          |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                          |
| OS-EXT-SRV-ATTR:instance_name       |                                               |
| OS-EXT-STS:power_state              | NOSTATE                                       |
| OS-EXT-STS:task_state               | scheduling                                    |
| OS-EXT-STS:vm_state                 | building                                      |
| OS-SRV-USG:launched_at              | None                                          |
| OS-SRV-USG:terminated_at            | None                                          |
| accessIPv4                          |                                               |
| accessIPv6                          |                                               |
| addresses                           |                                               |
| adminPass                           | 6Fa52DZ6AXEy                                  |
| config_drive                        |                                               |
| created                             | 2025-18-06T01:08:01Z                          |
| flavor                              | m1.tiny (1)                                   |
| hostId                              |                                               |
| id                                  | b942a0c6-7541-4380-91a4-19a16d875860          |
| image                               | cirros (1fa43af3-ac22-4dba-85e9-04b21807c5d4) |
| key_name                            | mykey                                         |
| name                                | demo1                                         |
| progress                            | 0                                             |
| project_id                          | 496d5b465e684f20a8217fa408728ac2              |
| properties                          |                                               |
| security_groups                     | name='default'                                |
| status                              | BUILD                                         |
| updated                             | 2025-18-06T01:08:02Z                          |
| user_id                             | 6714a667d1e3406fab0c7add7d71d412              |
| volumes_attached                    |                                               |
+-------------------------------------+-----------------------------------------------+
```

NOTE: configuration are located in /etc/kolla in all cluster nodes.

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@GPU111:/etc/kolla# ls -l
total 140
drwxrwx--- 2 root root 4096 Nov  5 11:44 cinder-api
drwxrwx--- 2 root root 4096 Nov  7 13:11 cinder-backup
drwxrwx--- 2 root root 4096 Nov  5 11:44 cinder-scheduler
drwxrwx--- 2 root root 4096 Nov  5 12:03 cinder-volume
drwxrwx--- 2 root root 4096 Nov  5 11:29 cron
drwxrwx--- 2 root root 4096 Nov  5 11:29 fluentd
drwxrwx--- 2 root root 4096 Nov  5 11:42 glance-api
drwxrwx--- 3 root root 4096 Nov  5 11:36 haproxy
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-api
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-api-cfn
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-engine
drwxrwx--- 2 root root 4096 Nov  5 12:25 horizon
drwxrwx--- 3 root root 4096 Nov  5 11:36 keepalived
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone-fernet
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone-ssh
drwxrwx--- 2 root root 4096 Nov  5 11:29 kolla-toolbox
drwxrwx--- 2 root root 4096 Nov  5 11:37 mariadb
drwxrwx--- 2 root root 4096 Nov  5 11:37 mariadb-clustercheck
drwxrwx--- 2 root root 4096 Nov  5 11:38 memcached
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-dhcp-agent
drwxrwx--- 2 root root 4096 Nov  5 12:21 neutron-l3-agent
drwxrwx--- 2 root root 4096 Nov  5 12:21 neutron-metadata-agent
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-openvswitch-agent
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-server
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-api
drwxrwx--- 2 root root 4096 Nov  5 12:05 nova-api-bootstrap
drwxrwx--- 2 root root 4096 Nov  5 12:06 nova-cell-bootstrap
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-conductor
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-novncproxy
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-scheduler
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-db-server
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-vswitchd
drwxrwx--- 2 root root 4096 Nov  5 12:05 placement-api
drwxrwx--- 2 root root 4096 Nov  5 11:39 rabbitmq
```

```
ubuntu@GPU71:~$ ls -al /etc/kolla/
total 44
drwxr-xr-x  11 root root 4096 Nov  5 12:20 .
drwxr-xr-x 107 root root 4096 Nov  5 12:07 ..
drwxrwx---   2 root root 4096 Nov  5 11:29 cron
drwxrwx---   2 root root 4096 Nov  5 11:29 fluentd
drwxrwx---   2 root root 4096 Nov  5 11:29 kolla-toolbox
drwxrwx---   2 root root 4096 Nov  5 12:20 neutron-openvswitch-agent
drwxrwx---   2 root root 4096 Nov  5 12:08 nova-compute
drwxrwx---   2 root root 4096 Nov  5 12:08 nova-libvirt
drwxrwx---   2 root root 4096 Nov  5 12:08 nova-ssh
drwxrwx---   2 root root 4096 Nov  5 12:18 openvswitch-db-server
drwxrwx---   2 root root 4096 Nov  5 12:18 openvswitch-vswitchd
```

Extra: Add Ceph storage as an Cinder backend

NOTE: Refer to https://verdysgn.io/8tkrVJP7QU-MjO-0_W13jQ for setup a Ceph storage.

This section provides steps to add a Ceph storage into Cinder volume as backend; and set Nova to use the Ceph storage for running instance.

Deploy OpenStack using Kolla-Ansible (Cons’t)

Setup Ceph storage via cephadm

tags: ceph
Example of Ceph storage cluster setup via cephadm.
I setup Ceph storage for OpenStack to use as "shared" storage which allow "live-migration" of instance.
OpenStack setup: https://verdysgn.io/v2HFpS2AS8CVXiv_I2svZg (kolla-ansible)

Environment

```
IP Address     hostname  Ceph Node
-------------------------------------------
192.168.1.23   GPU111   deploy & mon
192.168.1.24   GPU71    osd
192.168.1.25   GPU72    osd
192.168.1.26   GPU73    osd
```

The GPU111 will be deploy node, and initial ceph monitor host. (cephadm will auto add 2 addtional monitors automatically base on cluster size)The GPU71, GPU72 and GPU73 are OSD hosts, they have additional disk /dev/sdb (usb 3.0 drive) for OSD drive.

Steps

Create **cephadmin** user and set to sudoer

```
root@GPU111:~# useradd -m -s /bin/bash cephadmin
root@GPU111:~# passwd cephadmin
New password:
Retype new password:
passwd: password updated successfully
root@GPU111:~# echo "cephadmin ALL=(ALL:ALL) NOPASSWD:ALL" >> /etc/sudoers.d/cephadmin
root@GPU111:~# ls /etc/sudoers.d/
cephadmin  README
root@GPU111:~# chmod 0440 /etc/sudoers.d/cephadmin
```

Install **cephadm** tool

```
root@GPU111:~# apt install cephadm
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupdplugin1 libisns0
Use 'apt autoremove' to remove them.
Recommended packages:
  docker.io
The following NEW packages will be installed:
  cephadm
0 upgraded, 1 newly installed, 0 to remove and 11 not upgraded.
Need to get 58.5 kB of archives.
After this operation, 307 kB of additional disk space will be used.
Get:1 http://au.archive.ubuntu.com/ubuntu focal-updates/universe amd64 cephadm amd64 15.2.16-0ubuntu0.20.04.1 [58.5
kB]
Fetched 58.5 kB in 0s (665 kB/s)
Selecting previously unselected package cephadm.
(Reading database ... 130835 files and directories currently installed.)
Preparing to unpack .../cephadm_15.2.16-0ubuntu0.20.04.1_amd64.deb ...
Unpacking cephadm (15.2.16-0ubuntu0.20.04.1) ...
Setting up cephadm (15.2.16-0ubuntu0.20.04.1) ...
Adding system user cephadm....done
```

NOTE: or install via curl to fetch most recent version. For example

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@node1:~# curl --silent --remote-name --location https://github.com/ceph/ceph/raw/quincy/src/cephadm/cephadm
root@node1:~# chmod +x ./cephadm
```

Switch to **cephadmin** user

```
root@GPU111:~# su cephadmin -
cephadmin@GPU111:/root$
```

Bootstrap a new cluster

```
cephadmin@GPU111:~$ sudo cephadm bootstrap --mon-ip 192.168.1.23
Creating directory /etc/ceph for ceph.conf
Verifying podman|docker is present...
Verifying lvm2 is present...
Verifying time synchronization is in place...
Unit chrony.service is enabled and running
Repeating the final host check...
podman|docker (/usr/bin/docker) is present
systemctl is present
lvcreate is present
Unit chrony.service is enabled and running
Host looks OK
Cluster fsid: 16e21856-6030-11ed-8a24-db02e0e54b22
Verifying IP 192.168.1.23 port 3300 ...
Verifying IP 192.168.1.23 port 6789 ...
Mon IP 192.168.1.23 is in CIDR network 192.168.1.0/24
Pulling container image quay.io/ceph/ceph:v15...
Extracting ceph user uid/gid from container image...
Creating initial keys...
Creating initial monmap...
Creating mon...
Waiting for mon to start...
Waiting for mon...
mon is available
Assimilating anything we can from ceph.conf...
Generating new minimal ceph.conf...
```

NOTE: add --skip-monitoring-stack option to skip monitoring stack (Prometheus, Grafana, etc...) deploy.

NOTE: add --cluster-network to use cluster network. (for example, 192.168.122.0/24) (for newer cephadm version??) https://github.com/ceph/ceph/pull/38911

When login dashbaoard first time, you will be asked to update password. Here I simply update the password.

```
username/password: admin/cephadmin
```

Check Ceph containers

```
cephadmin@GPU111:~$ sudo docker ps
CONTAINER ID   IMAGE                                    COMMAND                   CREATED         STATUS         PORTS     NAMES
15225fbe7d8a   quay.io/prometheus/alertmanager:v0.20.0   "/bin/alertmanager -…"   3 minutes ago   Up 3 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-alertmanager.GPU111
23d37d214780   quay.io/ceph/ceph-grafana:6.7.4          "/bin/sh -c 'grafana…"   3 minutes ago   Up 3 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-grafana.GPU111
a506c9016de5   quay.io/prometheus/prometheus:v2.18.1     "/bin/prometheus --c…"   4 minutes ago   Up 4 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-prometheus.GPU111
795ec188ff2a   quay.io/prometheus/node-exporter:v0.18.1  "/bin/node_exporter …"   4 minutes ago   Up 4 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-node-exporter.GPU111
a497030e7c57   quay.io/ceph/ceph:v15                    "/usr/bin/ceph-crash…"   5 minutes ago   Up 5 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-crash.GPU111
19e955a4dadc   quay.io/ceph/ceph:v15                    "/usr/bin/ceph-mgr -…"   6 minutes ago   Up 6 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-mgr.GPU111.oezajx
6120dd3bf195   quay.io/ceph/ceph:v15                    "/usr/bin/ceph-mon -…"   6 minutes ago   Up 6 minutes
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-mon.GPU111
```

Deploy OpenStack using Kolla-Ansible (Cons’t)

Check Ceph status

```
cephadmin@GPU111:~$ sudo /usr/sbin/cephadm shell --fsid 16e21856-6030-11ed-8a24-db02e0e54b22 -c /etc/ceph/ceph.conf -k /etc/ceph/ceph.client.admin.keyring
Using recent ceph image quay.io/ceph/ceph@sha256:c08064dde4bba4e72a1f55d90ca32df9ef5aafab82efe2e0a0722444a5aaacca

root@GPU111:/# sudo ceph -s
  cluster:
    id:     16e21856-6030-11ed-8a24-db02e0e54b22
    health: HEALTH_WARN
            OSD count 0 < osd_pool_default_size 3

  services:
    mon: 1 daemons, quorum GPU111 (age 8m)
    mgr: GPU111.oezajx(active, since 7m)
    osd: 0 osds: 0 up, 0 in

  data:
    pools:   0 pools, 0 pgs
    objects: 0 objects, 0 B
    usage:   0 B used, 0 B / 0 B avail
    pgs:
```

```
cephadmin@GPU111:~$ sudo ceph -s
  cluster:
    id:     16e21856-6030-11ed-8a24-db02e0e54b22
    health: HEALTH_WARN
            OSD count 0 < osd_pool_default_size 3

  services:
    mon: 1 daemons, quorum GPU111 (age 12m)
    mgr: GPU111.oezajx(active, since 12m)
    osd: 0 osds: 0 up, 0 in

  data:
    pools:   0 pools, 0 pgs
    objects: 0 objects, 0 B
    usage:   0 B used, 0 B / 0 B avail
    pgs:
```

Install Ceph-common package so we can use ceph command to check cluster directly

```
cephadmin@GPU111:~$ sudo cephadm version
Using recent ceph image quay.io/ceph/ceph@sha256:c08064dde4bba4e72a1f55d90ca32df9ef5aafab82efe2e0a0722444a5aaacca
ceph version 15.2.17 (8a82819d84cf884bd39c17e3236e0632ac146dc4) octopus (stable)

cephadmin@GPU111:~$ sudo cephadm add-repo --release octopus
Installing repo GPG key from https://download.ceph.com/keys/release.asc...
Installing repo file at /etc/apt/sources.list.d/ceph.list...

cephadmin@GPU111:~$ sudo cephadm install ceph-common
Installing packages ['ceph-common']...
```

Add host label to GPU111

```
cephadmin@GPU111:~$ sudo ceph orch host ls
HOST    ADDR    LABELS  STATUS
GPU111  GPU111

cephadmin@GPU111:~$ sudo ceph orch host label add GPU111 _admin
Added label _admin to host GPU111

cephadmin@GPU111:~$ sudo ceph orch host ls
HOST    ADDR    LABELS  STATUS
GPU111  GPU111  _admin
```

Deploy OpenStack using Kolla-Ansible (Cons’t)

Copy public key of Ceph to OSD hosts

```
cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU71
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU71 (192.168.1.24)' can't be established.
ECDSA key fingerprint is SHA256:jwfxrwrwnLwLblfkfLJC3Yr894+FGQznY8+ylYTNXR8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU71's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@GPU71'"
and check to make sure that only the key(s) you wanted were added.

cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU72
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU72 (192.168.1.25)' can't be established.
ECDSA key fingerprint is SHA256:9gXSkvu9mZsfNu56cQgMcSYnfEr33h5haoYs8CVxh6E.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU72's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@GPU72'"
and check to make sure that only the key(s) you wanted were added.

cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU73
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU73 (192.168.1.26)' can't be established.
ECDSA key fingerprint is SHA256:4GMMsd+hO9Zh71ZrI/cHneRFoPO3kBuV/f5K7JeMVhY.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU73's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@GPU73'"
and check to make sure that only the key(s) you wanted were added.
```

Add OSD host to cluster and set label

```
cephadmin@GPU111:~$ sudo ceph orch host add GPU71
Added host 'GPU71'
cephadmin@GPU111:~$ sudo ceph orch host add GPU72
Added host 'GPU72'
cephadmin@GPU111:~$ sudo ceph orch host add GPU73
Added host 'GPU73'

cephadmin@GPU111:~$ sudo ceph orch host ls
HOST    ADDR    LABELS     STATUS
GPU111  GPU111  _admin mon
GPU71   GPU71
GPU72   GPU72
GPU73   GPU73

cephadmin@GPU111:~$ sudo ceph orch host label add GPU71 osd
Added label osd to host GPU71
cephadmin@GPU111:~$ sudo ceph orch host label add GPU72 osd
Added label osd to host GPU72
cephadmin@GPU111:~$ sudo ceph orch host label add GPU73 osd
Added label osd to host GPU73

cephadmin@GPU111:~$ sudo ceph orch host ls
HOST    ADDR    LABELS     STATUS
GPU111  GPU111  _admin mon
GPU71   GPU71   osd
GPU72   GPU72   osd
GPU73   GPU73   osd
```

NOTE: you could tell cephadm to deploy daemons base on label. For example, to limit monitors run on host with "mon" label.

Deploy OpenStack using Kolla-Ansible (Cons’t)

```
$ sudo ceph orch apply mon --placement="label:mon"
```

Add storage to cluster

```
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU71:/dev/sdb
Created osd(s) 0 on host 'GPU71'
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU72:/dev/sdb
Created osd(s) 1 on host 'GPU72'
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU73:/dev/sdb
Created osd(s) 2 on host 'GPU73'
```

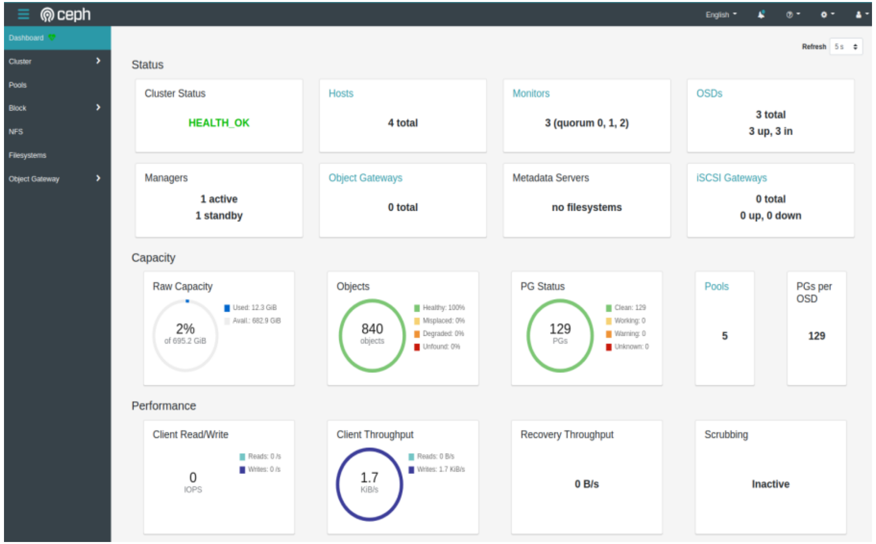
Check cluster status

```
cephadmin@GPU111:~$ sudo ceph -s
  cluster:
    id:     16e21856-6030-11ed-8a24-
db02e0e54b22
    health: HEALTH_OK

  services:
    mon: 3 daemons, quorum
GPU111,GPU71,GPU72 (age 119s)
    mgr: GPU111.oezajx(active, since 36m),
standbys: GPU71.pmlfuu
    osd: 3 osds: 3 up (since 24s), 3 in (since
24s)

  data:
    pools:   1 pools, 1 pgs
    objects: 0 objects, 0 B
    usage:   3.0 GiB used, 692 GiB / 695 GiB
avail
    pgs:     1 active+clean
```

Dasboard



Deploy OpenStack using Kolla-Ansible (Cons’t)

Purging cluster

```
root@GPU111:~# ceph mgr module disable cephadm

root@GPU111:~# ceph fsid
16e21856-6030-11ed-8a24-db02e0e54b22

cephadm rm-cluster --force --zap-osds --fsid 16e21856-6030-11ed-8a24-db02e0e54b22
```

To disable/remove the Prometheus rleated services (TBC)

```
$ ceph mgr module disable prometheus
$ ceph orch rm node-exporter
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Extra: create examples for demo

NOTE: copy "clouds.yaml" in controller node to /etc/kolla/ first.

Create pool for volume

```
root@GPU111:~# ceph osd pool create volumes
pool 'volumes' created
root@GPU111:~# ceph osd pool create vms
pool 'vms' created
root@GPU111:~# ceph osd pool create backups
pool 'backups' created
```

```
root@GPU111:~# rbd pool init volumes
root@GPU111:~# rbd pool init vms
root@GPU111:~# rbd pool init backups
```

NOTE: we still create "backups" pool because "Kolla-Ansible" playbook still add Ceph keyring for cinder-backup no matter "nfs" is configured to cinder_backup_driver. I think its kinda of bug. Fail message if no keyring for "cinder-backup" configured.

```
TASK [cinder : Copy over Ceph keyring files for cinder-backup] ****************************************************
An exception occurred during task execution. To see the full traceback, use -vvv. The error was: If you are using a module and expect the file to exist on the remote, see the remote_src option
failed: [GPU111] (item=cinder-backup/ceph.client.cinder.keyring) => {"ansible_loop_var": "item", "changed": false, "item": "cinder-backup/ceph.client.cinder.keyring", "msg": "Could not find or access '/etc/kolla/config/cinder/cinder-backup/ceph.client.cinder.keyring' on the Ansible Controller.\nIf you are using a module and expect the file to exist on the remote, see the remote_src option"}
An exception occurred during task execution. To see the full traceback, use -vvv. The error was: If you are using a module and expect the file to exist on the remote, see the remote_src option
failed: [GPU111] (item=cinder-backup/ceph.client.cinder-backup.keyring) => {"ansible_loop_var": "item", "changed": false, "item": "cinder-backup/ceph.client.cinder-backup.keyring", "msg": "Could not find or access '/etc/kolla/config/cinder/cinder-backup/ceph.client.cinder-backup.keyring' on the Ansible Controller.\nIf you are using a module and expect the file to exist on the remote, see the remote_src option"}
```

Setup Ceph client authentication to create cinder user for accessing volumes and vms pool.

```
root@GPU111:~# ceph auth get-or-create client.cinder mon 'profile rbd' osd 'profile rbd pool=volumes, profile rbd pool=vms' mgr 'profile rbd pool=volumes, profile rbd pool=vms'
[client.cinder]
        key = AQAC5nBjU4CrCRAAi3kUAheIpOz5dvWZKg9pNw==
```

Also for cinder-backup. (We still use NFS for cinder-backup anyway)

```
root@GPU111:/etc/kolla# ceph auth get-or-create client.cinder-backup mon 'profile rbd' osd 'profile rbd pool=backups' mgr 'profile rbd pool=backups'
[client.cinder-backup]
        key = AQB++HBjbprHIBAAai68wc04/vfhYJGmNgTQoQ==
```

Copy the key to deploy node for cinder

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/cinder/cinder-volume/

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/cinder-volume/ceph.client.cinder.keyring
[client.cinder]
        key = AQAC5nBjU4CrCRAAi3kUAheIpOz5dvWZKg9pNw==
```

Deploy OpenStack using Kolla-Ansible (Cons’t)

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/cinder/cinder-backup/

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/cinder-volume/ceph.client.cinder.keyring
./config/cinder/cinder-backup/

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/cinder-backup/ceph.client.cinder-backup.keyring
[client.cinder-backup]
        key = AQB++HBjbprHIBAAai68wc04/vfhYJGmNgTQoQ==
```

NOTE: cinder-backup needs cinder and cinder-backup keys.

Copy ceph.conf to deploy node

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/ceph.conf
# minimal ceph.conf for 16e21856-6030-11ed-8a24-db02e0e54b22
[global]
fsid = 16e21856-6030-11ed-8a24-db02e0e54b22
mon_host = [v2:192.168.1.23:3300/0,v1:192.168.1.23:6789/0]
```

NOTE: remove the "tab" in the ceph.conf. Otherwise kolla-ansible would fail with error.

```
TASK [cinder : Copying over ceph.conf for Cinder]

An exception occurred during task execution. To see the full traceback, use -vvv. The error was:
oslo_config.iniparser.ParseError: at line 3, Unexpected continuation line: '\tfsid = 16e21856-6030-11ed-8a24-db02e0e54b22'
fatal: [GPU111]: FAILED! => {"msg": "Unexpected failure during module execution.", "stdout": ""}
```

```
[global]
fsid =
mon_host =
```

Copy ceph.conf and cinder keyring for Nova

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/nova

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/ceph.conf ./config/nova/
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/cinder-volume/ceph.client.cinder.keyring ./config/nova/
```

Check Ceph config files for kolla

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ tree ./config/
./config/
├── cinder
│   ├── ceph.conf
│   ├── cinder-backup
│   │   ├── ceph.client.cinder-backup.keyring
│   │   └── ceph.client.cinder.keyring
│   └── cinder-volume
│       └── ceph.client.cinder.keyring
├── nfs_shares
└── nova
    ├── ceph.client.cinder.keyring
    └── ceph.conf
```

Config globals.yml

```
cinder_backend_ceph: "yes"
nova_backend_ceph: "yes"

ceph_cinder_keyring: "ceph.client.cinder.keyring"
ceph_cinder_user: "cinder"
ceph_cinder_pool_name: "volumes"
ceph_cinder_backup_keyring: "ceph.client.cinder-backup.keyring"
ceph_cinder_backup_user: "cinder-backup"
ceph_cinder_backup_pool_name: "backups"

ceph_nova_keyring: "{{ ceph_cinder_keyring }}"
ceph_nova_user: "{{ ceph_cinder_user }}"
ceph_nova_pool_name: "vms"
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Apply configuration

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i multinode deploy
```

Check cinder volume service status

```
root@GPU111:~# openstack volume service list
+------------------+--------------+------+---------+-------+----------------------------+
| Binary           | Host         | Zone | Status  | State | Updated At                 |
+------------------+--------------+------+---------+-------+----------------------------+
| cinder-scheduler | GPU111       | nova | enabled | up    | 2025-18-13T14:36:43.000000 |
| cinder-volume    | GPU111@nfs-1 | nova | enabled | up    | 2025-18-13T14:36:44.000000 |
| cinder-backup    | GPU111       | nova | enabled | up    | 2025-18-13T14:36:48.000000 |
| cinder-volume    | GPU111@rbd-1 | nova | enabled | up    | 2025-18-13T14:36:47.000000 |
+------------------+--------------+------+---------+-------+----------------------------+
```

The cinder-backup use NFS backup driver

```
root@GPU111:/etc/kolla/cinder-backup# cat cinder.conf  | grep backup
backup_driver = cinder.backup.drivers.nfs.NFSBackupDriver
backup_mount_options =
backup_mount_point_base = /var/lib/cinder/backup
backup_share = GPU111:/kolla_nfs
backup_file_size = 327680000
```

Cinder volume has two backends enabled

```
root@GPU111:/etc/kolla/cinder-volume# cat cinder.conf | grep enabled_backends
enabled_backends = rbd-1,nfs-1
```

```
[rbd-1]
volume_driver = cinder.volume.drivers.rbd.RBDDriver
volume_backend_name = rbd-1
rbd_pool = volumes
rbd_ceph_conf = /etc/ceph/ceph.conf
rados_connect_timeout = 5
rbd_user = cinder
rbd_secret_uuid = a8c5fb21-a2da-46a3-9275-2c3365bc3a12
report_discard_supported = True

[nfs-1]
volume_driver = cinder.volume.drivers.nfs.NfsDriver
volume_backend_name = nfs-1
nfs_shares_config = /etc/cinder/nfs_shares
nfs_snapshot_support = True
nas_secure_file_permissions = False
nas_secure_file_operations = False
```

Create volume types

```
root@GPU111:/etc/kolla/cinder-volume# openstack volume type create ceph
+-------------+--------------------------------------+
| Field       | Value                                |
+-------------+--------------------------------------+
| description | None                                 |
| id          | f2c1ce62-d33f-4045-b94b-fce501961679 |
| is_public   | True                                 |
| name        | ceph                                 |
+-------------+--------------------------------------+

root@GPU111:/etc/kolla/cinder-volume# openstack volume type set ceph --property volume_backend_name=rbd-1
```

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@GPU111:/etc/kolla/cinder-volume# openstack volume type list --long
+--------------------------------------+-------------+-----------+---------------------+------------------------------+
| ID                                   | Name        | Is Public | Description         | Properties                   |
+--------------------------------------+-------------+-----------+---------------------+------------------------------+
| f2c1ce62-d33f-4045-b94b-fce501961679 | ceph        | True      | None                | volume_backend_name='rbd-1'  |
| f9782ab2-931d-430b-92dd-b6810677c0e0 | __DEFAULT__ | True      | Default Volume Type |                              |
+--------------------------------------+-------------+-----------+---------------------+------------------------------+
```

Test VM creation with ceph volume type

```
root@GPU111:/etc/kolla/cinder-volume# openstack flavor list
+--------------------------------------+-----------+-------+------+-----------+-------+-----------+
| ID                                   | Name      |   RAM | Disk | Ephemeral | VCPUs | Is Public |
+--------------------------------------+-----------+-------+------+-----------+-------+-----------+
| 1                                    | m1.tiny   |   512 |    1 |         0 |     1 | True      |
| 2                                    | m1.small  |  2048 |   20 |         0 |     1 | True      |
| 3                                    | m1.medium |  4096 |   40 |         0 |     2 | True      |
| 4                                    | m1.large  |  8192 |   80 |         0 |     4 | True      |
| 5                                    | m1.xlarge | 16384 |  160 |         0 |     8 | True      |
| 8f99ab4a-e2de-493d-9940-37de21dfd623 | 1c2r4d    |  2048 |    4 |         0 |     1 | False     |
+--------------------------------------+-----------+-------+------+-----------+-------+-----------+

root@GPU111:/etc/kolla/cinder-volume# openstack keypair list
+-----------+-------------------------------------------------+
| Name      | Fingerprint                                     |
+-----------+-------------------------------------------------+
| key-pair1 | 31:a1:03:89:15:fb:5b:46:e7:b8:5a:94:8d:39:82:12 |
+-----------+-------------------------------------------------+

root@GPU111:/etc/kolla/cinder-volume# openstack image list
+--------------------------------------+-------------+--------+
| ID                                   | Name        | Status |
+--------------------------------------+-------------+--------+
| 1fa43af3-ac22-4dba-85e9-04b21807c5d4 | cirros      | active |
| 40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4 | ubuntu22.04 | active |
+--------------------------------------+-------------+--------+

root@GPU111:/etc/kolla/cinder-volume# openstack network list
+--------------------------------------+----------+--------------------------------------+
| ID                                   | Name     | Subnets                              |
+--------------------------------------+----------+--------------------------------------+
| 17cb537a-88ac-42ed-9eb2-29177cb7490e | public1  | 142f17ca-c06d-4c07-abdc-dc935b1b83a3 |
| 67e89302-e5c7-4a17-ad1c-675b664874c1 | demo-net | 312eff74-d80c-4281-a51a-9b4bed5132c4 |
+--------------------------------------+----------+--------------------------------------+
```

Create a VM with "ephemeral" disk.

```
root@GPU111:~# openstack server create --flavor 1c2r4d --image ubuntu22.04 --key-name key-pair1 --network demo-net
ubuntu_vm
+-------------------------------------+----------------------------------------------------+
| Field                               | Value                                              |
+-------------------------------------+----------------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                             |
| OS-EXT-AZ:availability_zone         |                                                    |
| OS-EXT-SRV-ATTR:host                | None                                               |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                               |
| OS-EXT-SRV-ATTR:instance_name       |                                                    |
| OS-EXT-STS:power_state              | NOSTATE                                            |
| OS-EXT-STS:task_state               | scheduling                                         |
| OS-EXT-STS:vm_state                 | building                                           |
| OS-SRV-USG:launched_at              | None                                               |
| OS-SRV-USG:terminated_at            | None                                               |
| accessIPv4                          |                                                    |
| accessIPv6                          |                                                    |
| addresses                           |                                                    |
| adminPass                           | s5vdJaDqsBh7                                       |
| config_drive                        |                                                    |
| created                             | 2025-18-14T11:58:01Z                               |
| flavor                              | 1c2r4d (8f99ab4a-e2de-493d-9940-37de21dfd623)      |
| hostId                              |                                                    |
| id                                  | 1adcbf1f-0dd9-4086-a67f-ccf73d8b94e6               |
| image                               | ubuntu22.04 (40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4) |
| key_name                            | key-pair1                                          |
| name                                | ubuntu_vm                                          |
| progress                            | 0                                                  |
| project_id                          | 496d5b465e684f20a8217fa408728ac2                   |
| properties                          |                                                    |
| security_groups                     | name='default'                                     |
| status                              | BUILD                                              |
| updated                             | 2025-18-14T11:58:01Z                               |
| user_id                             | 6714a667d1e3406fab0c7add7d71d412                   |
| volumes_attached                    |                                                    |
+-------------------------------------+----------------------------------------------------+

root@GPU111:~# openstack server show ubuntu_vm
+-------------------------------------+----------------------------------------------------+
| Field                               | Value                                              |
+-------------------------------------+----------------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                             |
| OS-EXT-AZ:availability_zone         | nova                                               |
| OS-EXT-SRV-ATTR:host                | GPU72                                              |
| OS-EXT-SRV-ATTR:hypervisor_hostname | GPU7CJYHN2                                         |
| OS-EXT-SRV-ATTR:instance_name       | instance-00000007                                  |
| OS-EXT-STS:power_state              | Running                                            |
| OS-EXT-STS:task_state               | None                                               |
| OS-EXT-STS:vm_state                 | active                                             |
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Test live migration

```
root@GPU111:~# openstack server migrate --live-migration ubuntu_vm
root@GPU111:~# openstack server show ubuntu_vm
+-------------------------------------+----------------------------------------------------------+
| Field                               | Value                                                    |
+-------------------------------------+----------------------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                                   |
| OS-EXT-AZ:availability_zone         | nova                                                     |
| OS-EXT-SRV-ATTR:host                | GPU71                                                    |
| OS-EXT-SRV-ATTR:hypervisor_hostname | GPU7CJYHN1                                               |
| OS-EXT-SRV-ATTR:instance_name       | instance-00000007                                        |
| OS-EXT-STS:power_state              | Running                                                  |
| OS-EXT-STS:task_state               | None                                                     |
| OS-EXT-STS:vm_state                 | active                                                   |
| OS-SRV-USG:launched_at              | 2025-18-14T11:59:04.000000                               |
| OS-SRV-USG:terminated_at            | None                                                     |
| accessIPv4                          |                                                          |
| accessIPv6                          |                                                          |
| addresses                           | demo-net=10.0.0.142                                      |
| config_drive                        |                                                          |
| created                             | 2025-18-14T11:58:01Z                                     |
| flavor                              | 1c2r4d (8f99ab4a-e2de-493d-9940-37de21dfd623)            |
| hostId                              | c7db1b2553c921575d60036b5d792993c9a242db70405cb52bc0bfea |
| id                                  | 1adcbf1f-0dd9-4086-a67f-ccf73d8b94e6                     |
| image                               | ubuntu22.04 (40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4)       |
| key_name                            | key-pair1                                                |
| name                                | ubuntu_vm                                                |
| progress                            | 0                                                        |
| project_id                          | 496d5b465e684f20a8217fa408728ac2                         |
| properties                          |                                                          |
| security_groups                     | name='default'                                           |
| status                              | ACTIVE                                                   |
| updated                             | 2025-18-14T12:33:46Z                                     |
| user_id                             | 6714a667d1e3406fab0c7add7d71d412                         |
| volumes_attached                    |                                                          |
+-------------------------------------+----------------------------------------------------------+
```

Create cinder volume on ceph

```
root@GPU111:~# openstack volume create --bootable --image ubuntu22.04 --type ceph --size 3 ubuntu22.04_boot_image
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| attachments         | []                                   |
| availability_zone   | nova                                 |
| bootable            | false                                |
| consistencygroup_id | None                                 |
| created_at          | 2025-18-14T12:37:31.146975           |
| description         | None                                 |
| encrypted           | False                                |
| id                  | 6c4f5912-4480-46cd-a960-f0a59a3f5d98 |
| migration_status    | None                                 |
| multiattach         | False                                |
| name                | ubuntu22.04_boot_image               |
| properties          |                                      |
| replication_status  | None                                 |
| size                | 3                                    |
| snapshot_id         | None                                 |
| source_volid        | None                                 |
| status              | creating                             |
| type                | ceph                                 |
| updated_at          | None                                 |
| user_id             | 6714a667d1e3406fab0c7add7d71d412     |
+---------------------+--------------------------------------+

root@GPU111:~# openstack volume list --long
+--------------------------------------+------------------------+-----------+------+------+----------+-------------+------------+
| ID                                   | Name                   | Status    | Size | Type | Bootable | Attached to | Properties |
+--------------------------------------+------------------------+-----------+------+------+----------+-------------+------------+
| 6c4f5912-4480-46cd-a960-f0a59a3f5d98 | ubuntu22.04_boot_image | available |    3 | ceph | true     |             |            |
+--------------------------------------+------------------------+-----------+------+------+----------+-------------+------------+

root@GPU111:~# rbd ls -l volumes
NAME                                        SIZE   PARENT  FMT  PROT  LOCK
volume-6c4f5912-4480-46cd-a960-f0a59a3f5d98  3 GiB          2
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Create VM with the volume

```
root@GPU111:~# openstack server create --volume ubuntu22.04_boot_image --flavor 1c2r4d --key-name key-pair1 --network demo-net ubuntu22.04_ceph_volume_vm
+-------------------------------------+-----------------------------------------------+
| Field                               | Value                                         |
+-------------------------------------+-----------------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                        |
| OS-EXT-AZ:availability_zone         |                                               |
| OS-EXT-SRV-ATTR:host                | None                                          |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                          |
| OS-EXT-SRV-ATTR:instance_name       |                                               |
| OS-EXT-STS:power_state              | NOSTATE                                       |
| OS-EXT-STS:task_state               | scheduling                                    |
| OS-EXT-STS:vm_state                 | building                                      |
| OS-SRV-USG:launched_at              | None                                          |
| OS-SRV-USG:terminated_at            | None                                          |
| accessIPv4                          |                                               |
| accessIPv6                          |                                               |
| addresses                           |                                               |
| adminPass                           | vTNp7WWrhuFi                                  |
| config_drive                        |                                               |
| created                             | 2025-18-14T12:41:40Z                          |
| flavor                              | 1c2r4d (8f99ab4a-e2de-493d-9940-37de21dfd623) |
| hostId                              |                                               |
| id                                  | b7143bbd-f23e-42f9-9125-0b61592602f0          |
| image                               |                                               |
| key_name                            | key-pair1                                     |
| name                                | ubuntu22.04_ceph_volume_vm                    |
| progress                            | 0                                             |
| project_id                          | 496d5b465e684f20a8217fa408728ac2              |
| properties                          |                                               |
| security_groups                     | name='default'                                |
| status                              | BUILD                                         |
| updated                             | 2025-18-14T12:41:40Z                          |
| user_id                             | 6714a667d1e3406fab0c7add7d71d412              |
| volumes_attached                    |                                               |
+-------------------------------------+-----------------------------------------------+

root@GPU111:~# openstack server list
+--------------------------------------+----------------------------+--------+---------------------+-------------+--------+
| ID                                   | Name                       | Status | Networks            | Image       | Flavor |
+--------------------------------------+----------------------------+--------+---------------------+-------------+--------+
| b7143bbd-f23e-42f9-9125-0b61592602f0 | ubuntu22.04_ceph_volume_vm | ACTIVE | demo-net=10.0.0.182 |             | 1c2r4d |
| 1adcbf1f-0dd9-4086-a67f-ccf73d8b94e6 | ubuntu_vm                  | ACTIVE | demo-net=10.0.0.142 | ubuntu22.04 | 1c2r4d |
+--------------------------------------+----------------------------+--------+---------------------+-------------+--------+
```

Extra: Use local docker registry to deploy (Not Work)

Create a local docker registry on host GPU10 for Kolla-Ansible

NOTE: This does not work because kolla-ansible use "quay.io" to pull images and under "openstack.kolla" organization. However the docker registry **https://registry-1.docker.io** does not have "openstack.kolla". This results image pull failure.

Error message exampes:

```
root@GPU111:~# docker pull GPU10:5000/openstack.kolla/fluentd:master-rocky-9
Error response from daemon: manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown
```

```
docker.errors.NotFound: 404 Client Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd: Not Found (\"manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Create /etc/registry directory

```
root@GPU10:~# mkdir -p /etc/docker/registry
root@GPU10:~# cd /etc/docker/registry
root@GPU10:/etc/docker/registry#
```

Create **docker-compose.yml** file

```
root@GPU10:/etc/docker/registry# cat docker-compose.yml
version: '3'

services:
  registry:
    container_name: registry
    image: registry:2
    ports:
    - "5000:5000"
    environment:
      REGISTRY_STORAGE_FILESYSTEM_ROOTDIRECTORY: /data
    volumes:
      - /mnt/docker-registry-data:/data
      - /etc/docker/registry/config.yml:/etc/docker/registry/config.yml
```

Create config.yml file

```
root@GPU10:/etc/docker/registry# cat ./config.yml
version: 0.1
log:
  fields:
    service: registry
storage:
  cache:
    blobdescriptor: inmemory
  filesystem:
    rootdirectory: /var/lib/registry
http:
  addr: :5000
  headers:
    X-Content-Type-Options: [nosniff]
health:
  storagedriver:
    enabled: true
    interval: 10s
    threshold: 3
proxy:
  remoteurl: https://registry-1.docker.io
```

Run docker registry container

```
root@GPU10:/etc/docker/registry# docker-compose up -d
Creating network "registry_default" with the default driver
Pulling registry (registry:2)...
2: Pulling from library/registry
ca7dd9ec2225: Pull complete
c41ae7ad2b39: Pull complete
1ed0fc8a6161: Pull complete
21df229223d2: Pull complete
626897ccab21: Pull complete
Digest: sha256:ce14a6258f37702ff3cd92232a6f5b81ace542d9f1631966999e9f7c1ee6ddba
Status: Downloaded newer image for registry:2
Creating registry ... done
```

Configure insecure-registries and registry-mirrors

```
root@GPU10:/etc/docker/registry# cat /etc/docker/daemon.json
{
  "insecure-registries": ["GPU10:5000"],
  "registry-mirrors": ["http://GPU10:5000"]
}

root@GPU10:/etc/docker/registry# systemctl restart docker
```

Test docker registry

```
root@GPU10:/etc/docker/registry# docker pull fedora
Using default tag: latest
latest: Pulling from library/fedora
cb8b1ed77979: Pull complete
Digest: sha256:f99efcddc4dd6736d8a88cc1ab6722098ec1d77dbf7aed9a7a514fc997ca08e0
Status: Downloaded newer image for fedora:latest
docker.io/library/fedora:latest
```

```
root@GPU10:/etc/docker/registry# du -sh /mnt/docker-registry-data/docker/registry/v2/
64M                         /mnt/docker-registry-data/docker/registry/v2/
```

NOTE: docker will query configured registry mirror which trigger mirror download image. You should see image data usage increased in "/mnt/docker-registry-data/docker/registry/v2/blobs/")

Deploy OpenStack using Kolla-Ansible (Cons't)

If test success, configure **/etc/docker/daemon.json** on all OpenStack nodes. For example on controller host:

```
ubuntu@GPU111:~$ cat /etc/docker/daemon.json
{
   "bridge": "none",
   "ip-forward": false,
   "iptables": false,
   "insecure-registries" : ["GPU10:5000"],
   "registry-mirrors": ["http://GPU10:5000"]
}

ubuntu@GPU111:~$ sudo systemctl restart docker
```

Edit globals.yml

```
docker_registry: GPU10:5000
docker_registry_insecure: "yes"
```

Run kolla-ansible

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i multinode deploy
```

NOTE: Do not run kolla-ansible precheck, it removes "registry-mirrors" in docker daemon config.

NOTE: We have to set "insecure-registries" because we do not use https.

Troubleshoot

Error when use local docker registry.

```
RUNNING HANDLER [common : Restart fluentd container]
*******************************************************************************************
fatal: [GPU111]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n  File \"/usr/local/lib/python3.8/dist-packages/docker/api/client.py\", line 268, in _raise_for_status\\n    response.raise_for_status()\\n  File \"/usr/lib/python3/dist-packages/requests/models.py\", line 940, in raise_for_status\\n    raise HTTPError(http_error_msg, response=self)\\nrequests.exceptions.HTTPError: 500 Server Error: Internal Server Error for url: http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring handling of the above exception, another exception occurred:\\n\\nTraceback (most recent call last):\\n  File \"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\", line 381, in main\\n  File \"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 660, in recreate_or_restart_container\\n    self.pull_image()\\n  File \"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 450, in pull_image\\n    json.loads(line.strip().decode(\\'utf-8\\')) for line in self.dc.pull(\\n  File \"/usr/local/lib/python3.8/dist-packages/docker/api/image.py\", line 430, in pull\\n    self._raise_for_status(response)\\n  File \"/usr/local/lib/python3.8/dist-packages/docker/api/client.py\", line 270, in _raise_for_status\\n    raise create_api_error_from_http_exception(e)\\n  File \"/usr/local/lib/python3.8/dist-packages/docker/errors.py\", line 31, in create_api_error_from_http_exception\\n    raise cls(e, response=response, explanation=explanation)\\ndocker.errors.APIError: 500 Server Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd: Internal Server Error (\"Get \"https://GPU10:4000/v2/\": http: server gave HTTP response to HTTPS client\")\\n'"}
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Troubleshoot

Error when use local docker registry.

```
fatal: [GPU72]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/client.py\", line 268, in _raise_for_status\\n    response.raise_for_status()\\n  File \"/usr/lib/python3/dist-
packages/requests/models.py\", line 940, in raise_for_status\\n    raise HTTPError(http_error_msg,
response=self)\\nrequests.exceptions.HTTPError: 500 Server Error: Internal Server Error for url:
http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring
handling of the above exception, another exception occurred:\\n\\nTraceback (most recent call last):\\n  File
\"/tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\", line 381, in main\\n  File
\"/tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 660, in
recreate_or_restart_container\\n    self.pull_image()\\n  File
\"/tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 450, in
pull_image\\n    json.loads(line.strip().decode(\\'utf-8\\')) for line in self.dc.pull(\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/image.py\", line 430, in pull\\n    self._raise_for_status(response)\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/client.py\", line 270, in _raise_for_status\\n    raise create_api_error_from_http_exception(e)\\n  File
\"/usr/local/lib/python3.8/dist-packages/docker/errors.py\", line 31, in create_api_error_from_http_exception\\n    raise cls(e,
response=response, explanation=explanation)\\ndocker.errors.APIError: 500 Server Error for
http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd: Internal Server
Error (\"Get \"https://GPU10:4000/v2/\": x509: certificate relies on legacy Common Name field, use SANs instead\")\\n'"}
```

```
fatal: [GPU111]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/client.py\", line 268, in _raise_for_status\\n    response.raise_for_status()\\n  File \"/usr/lib/python3/dist-
packages/requests/models.py\", line 940, in raise_for_status\\n    raise HTTPError(http_error_msg,
response=self)\\nrequests.exceptions.HTTPError: 404 Client Error: Not Found for url: http+docker://localhost/v1.41/images/create?tag=master-
rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring handling of the above exception, another exception
occurred:\\n\\nTraceback (most recent call last):\\n  File
\"/tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\", line 381, in main\\n  File
\"/tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 651, in
recreate_or_restart_container\\n    self.start_container()\\n  File
\"/tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 669, in
start_container\\n    self.pull_image()\\n  File
\"/tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\", line 450, in
pull_image\\n    json.loads(line.strip().decode(\\'utf-8\\')) for line in self.dc.pull(\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/image.py\", line 430, in pull\\n    self._raise_for_status(response)\\n  File \"/usr/local/lib/python3.8/dist-
packages/docker/api/client.py\", line 270, in _raise_for_status\\n    raise create_api_error_from_http_exception(e)\\n  File
\"/usr/local/lib/python3.8/dist-packages/docker/errors.py\", line 31, in create_api_error_from_http_exception\\n    raise cls(e,
response=response, explanation=explanation)\\ndocker.errors.NotFound: 404 Client Error for
http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd: Not Found
(\"manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown\")\\n'"}
```


Troubleshoot

Error when use local docker registry.

kolla-ansible -i INVENTORY deploy is used to deploy and start all Kolla containers.

kolla-ansible -i INVENTORY destroy is used to clean up containers and volumes in the cluster.

kolla-ansible -i INVENTORY mariadb_recovery is used to recover a completely stopped mariadb cluster.

kolla-ansible -i INVENTORY prechecks is used to check if all requirements are meet before deploy for each of the OpenStack services.

kolla-ansible -i INVENTORY post-deploy is used to do post deploy on deploy node to get the admin openrc file.

kolla-ansible -i INVENTORY pull is used to pull all images for containers.

kolla-ansible -i INVENTORY reconfigure is used to reconfigure OpenStack service.

kolla-ansible -i INVENTORY upgrade is used to upgrades existing OpenStack Environment.

kolla-ansible -i INVENTORY stop is used to stop running containers.

kolla-ansible -i INVENTORY deploy-containers is used to check and if necessary update containers, without generating configuration.

kolla-ansible -i INVENTORY prune-images is used to prune orphaned Docker images on hosts.

kolla-ansible -i INVENTORY genconfig is used to generate configuration files for enabled OpenStack services, without then restarting the containers so it is not applied right away.

kolla-ansible -i INVENTORY1 -i INVENTORY2 ... Multiple inventories can be specified by passing the --inventory or -i command line option multiple times. This can be useful to share configuration between multiple environments. Any common configuration can be set in INVENTORY1 and INVENTORY2 can be used to set environment specific details.

kolla-ansible -i INVENTORY gather-facts is used to gather Ansible facts, for example to populate a fact cache.

Mauliate Godang